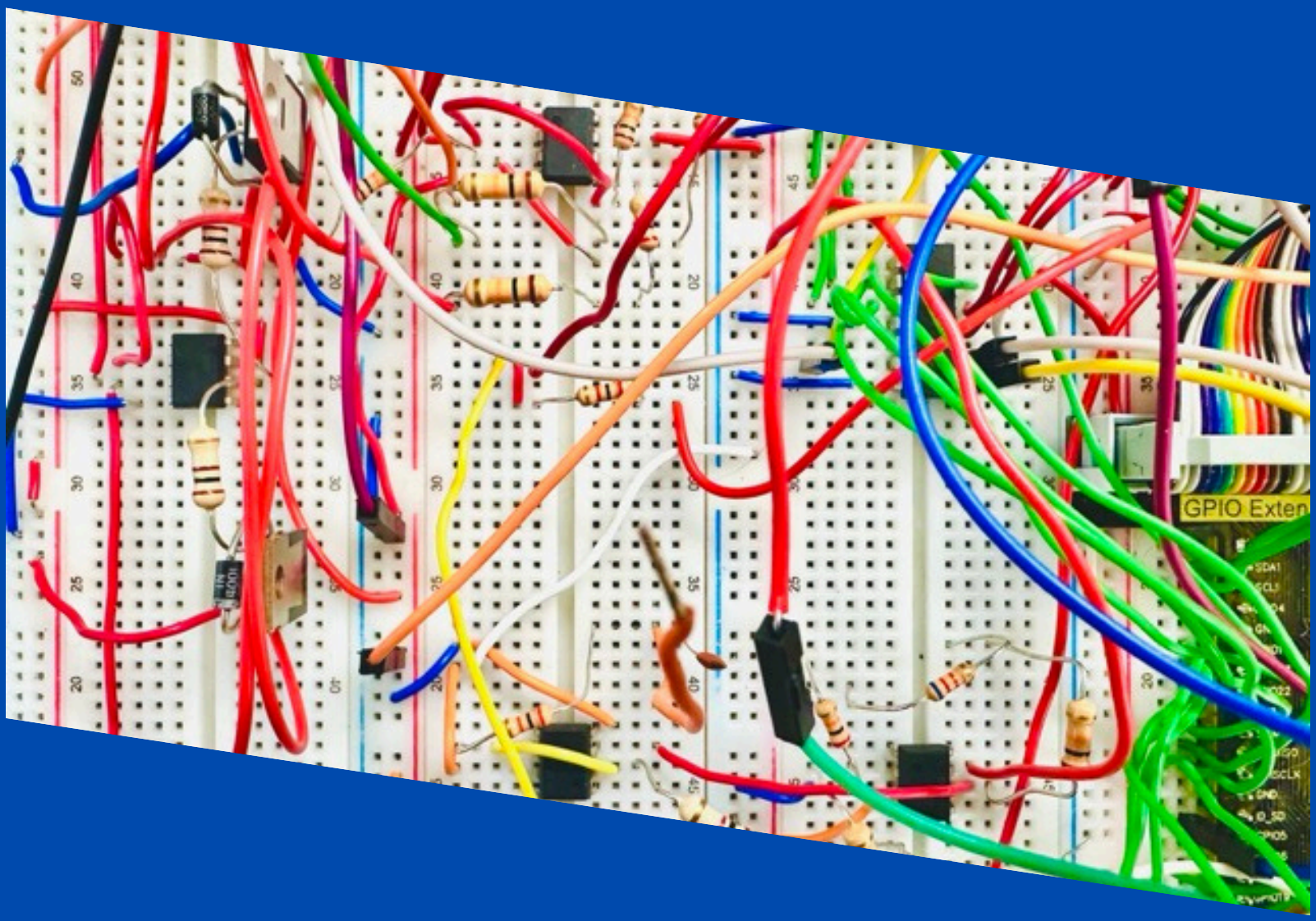
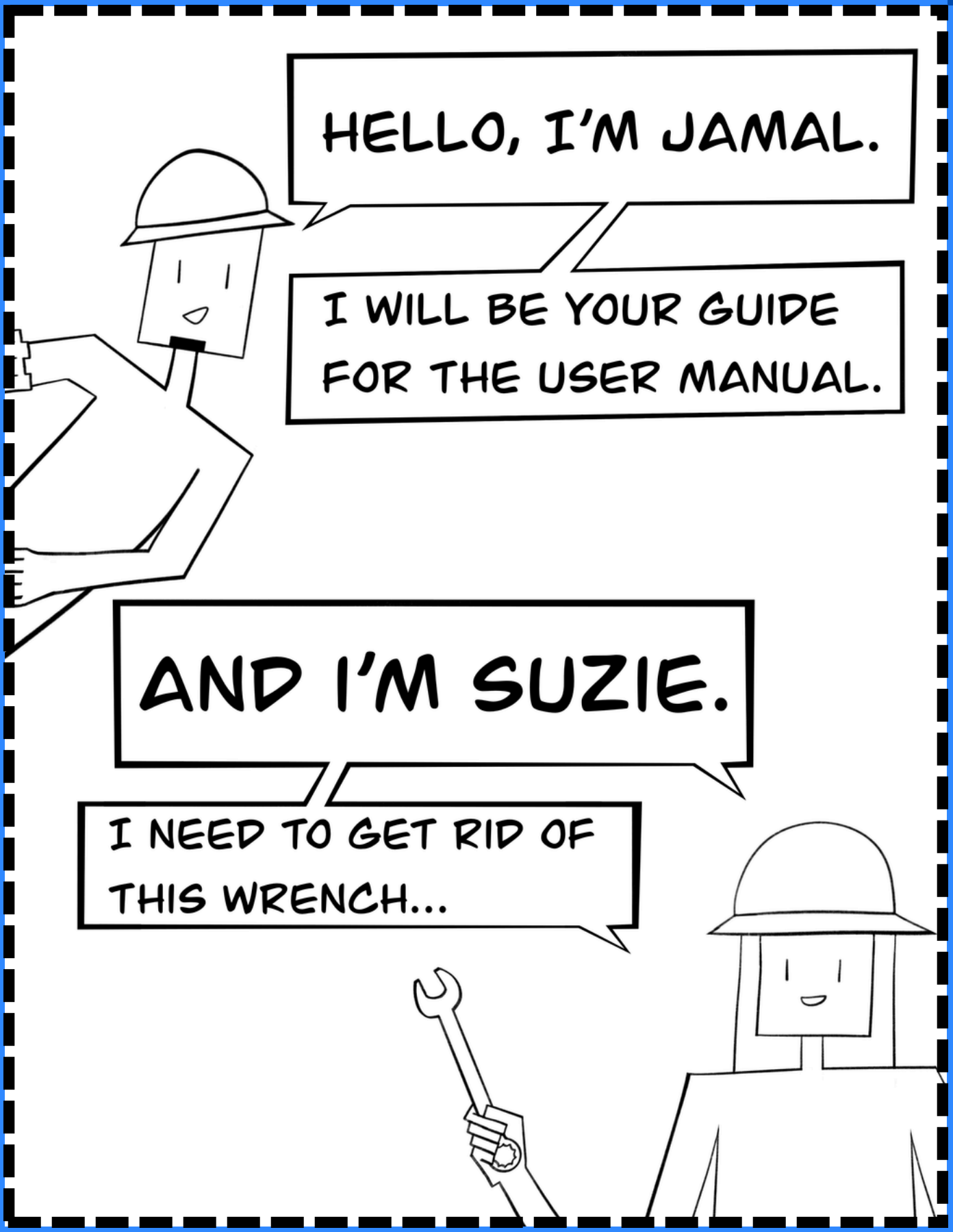
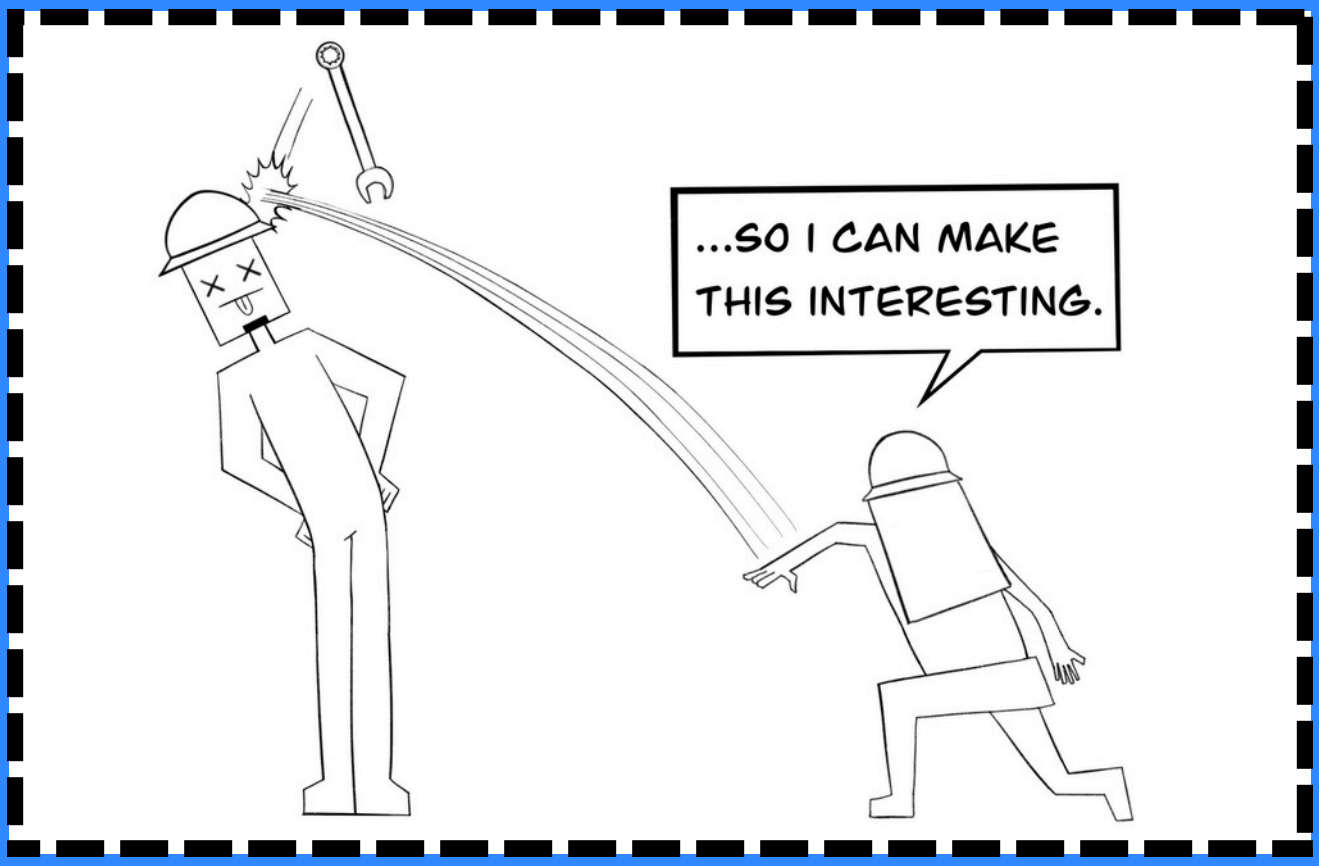
Group 5

# ELECTRONIC TOOL BENCH

USER MANUAL
(CONTAINS HEALTH AND SAFETY INFORMATION)




HELLO, I'M JAMAL.
I WILL BE YOUR GUIDE FOR THE USER MANUAL.
AND I'M SUZIE.
I NEED TO GET RID OF THIS WRENCH...




**⚠WARNING** JAMAL AND SUZIE ARE SMART BUT THEY ARE OBSESSED WITH CARTOON VIOLENCE...

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# HEALTH AND SAFETY

# HEALTH AND SAFETY

PLEASE READ THIS MANUAL CAREFULLY BEFORE PROCEEDING TO USE THE ELECTRONICS TOOL BENCH SYSTEM. FAILING TO SO DO MAY CAUSE INJURY OR DAMAGE THE SYSTEM.

AS YOU READ THIS MANUAL, YOU WILL ENCOUNTER THESE LABELS WITH INFORMATION ON USING THE ELECTRONIC TOOL BENCH SYSTEM PROPERLY AND SAFELY.

**⚠ CAUTION**

CAUTIONS YOU ABOUT THE INCORRECT USE OF THE ELECTRONICS TOOL BENCH THAT MAY RESULT IN MILD OR MODERATE INJURY.

**⚠ WARNING**

WARNS YOU ABOUT THE INCORRECT USE OF THE ELECTRONICS TOOL BENCH THAT MAY RESULT IN SERIOUS INJURY AND/OR DAMAGE TO THE ELECTRONICS TOOL BENCH SYSTEM OR ITS COMPONENTS.

## ⚠WARNING OPERATION

- DO NOT USE THE ELECTRONICS TOOL BENCH ON AN UNEVEN OR INCLINED SURFACE. INSTEAD, SET IT ON A LEVEL AND FLAT SURFACE.
- ONLY OPERATE THE SYSTEM IN A HEAT-REGULATED ROOM OR AT CONDITIONS ANALOGOUS TO A LABORATORY ENVIRONMENT.
- DO NOT PLUG ANY CIRCUIT PARTS UNLESS THE USER MANUAL EXPLICITLY STATES THAT IT IS ACCEPTABLE.
- CHECK IF THE 24V POWER SUPPLY IS CONNECTED TO THE ELECTRONICS TOOL BENCH SYSTEM. GO TO PAGE 8 FOR MORE INFORMATION.
- DO NOT USE THE ELECTRONIC TOOL BENCH DURING A LIGHTNING STORM. THERE MAY BE A RISK OF ELECTRIC SHOCK.

- KEEP FOOD AND DRINKS AWAY FROM THE ELECTRONICS TOOL BENCH SYSTEM.

## ⚠WARNING FIRE

IF THE ELECTRICAL TOOL BENCH CATCHES FIRE, THEN UNPLUG THE POWER SUPPLY IMMEDIATELY. THEN, EXTINGUISH THE FIRE WITH EITHER:

- CLASS C FIRE EXTINGUISHER
- THICK BLANKET OR TOWEL
- BAKING SODA

IF YOU ARE UNABLE TO EXTINGUISH THE FIRE, THEN GET OUT OF THE ROOM AND CLOSE THE DOOR. CALL 911 IMMEDIATELY AT A SAFE DISTANCE FROM THE FIRE.

DO NOT POUR WATER OR OTHER LIQUID INTO THE FIRE.

## ⚠WARNING POWER SUPPLY

- USE THE OFFICIAL 24VDC RASPBERRY PI 4 POWER SUPPLY. DO NOT USE POWER SOURCES OR ADAPTERS FROM OTHER VENDORS.
- DO NOT PLUG THE POWER SUPPLY TO ANY DEVICE THAT IS NOT THE ELECTRONICS TOOL BENCH SYSTEM.

## ⚠ CAUTION LCD

- THE LCD USES A STRONG BACKLIGHT TO DISTINGUISH THE TEXT FROM THE BLUE SCREEN.
- IF THE SCREEN CAUSES BOTHERING ON YOUR EYES, THEN MOVE YOUR EYES AS FAR AWAY FROM THE SCREEN AS POSSIBLE.
- DO NOT WEAR SUNGLASSES OR SHADES WHILE OPERATING THE ELECTRONIC TOOL BENCH SYSTEM. YOU MAY MAKE INCORRECT CONNECTIONS OF EXTERNAL CIRCUIT ELEMENTS ONTO THE SYSTEM.

IF YOU EXPERIENCE DIZZINESS, FATIGUE, OR NAUSEA WHILE USING THE ELECTRONIC TOOL BENCH SYSTEM, POWER IT OFF IMMEDIATELY.

## ⚠ CAUTION TRANSPORTATION

- TO KEEP THE ELECTRONIC TOOL BENCH INTACT WHILE TRANSPORTING IT FROM ONE PLACE TO ANOTHER, HAVE ANOTHER PERSON ASSIST IN CARRYING IT.

# 0.SETTING THE SYSTEM

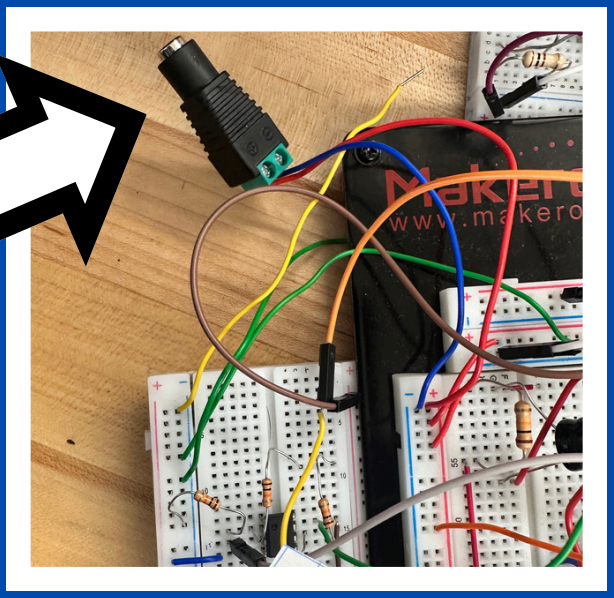
# SETTING THE SYSTEM

**⚠WARNING** DO NOT USE THE ELECTRONIC TOOL BENCH DURING A LIGHTNING STORM.THERE MAY BE A RISK OF ELECTRIC SHOCK.

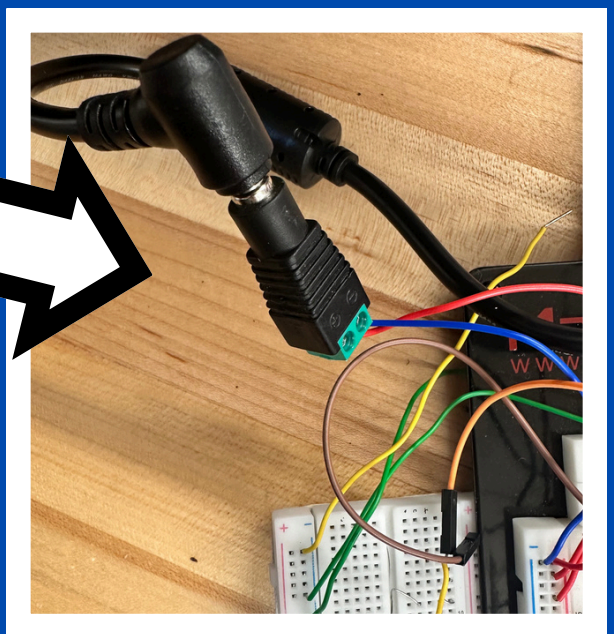
**⚠WARNING** PLEASE USE THE OFFICIAL 24VDC RASPBERRY PI POWER SUPPLY. DO NOT USE POWER ADAPTERS FROM OTHER VENDORS.

## CONNECTING THE SYSTEM

IN THE TOP LEFT CORNER OF THE ELECTRONICS TOOL BENCH, YOU WILL SEE A POWER SOCKET.



CONNECT THE 24VDC RASPBERRY PI POWER SUPPLY TO THE POWER SOCKET.



**⚠WARNING**

DO NOT USE THE 5V OR 3V PINS OF THE RASPBERRY PI 4 TO CONNECT THE POWER. DO NOT TOUCH ANY OF THE RASPBERRY PI 4 PINS.

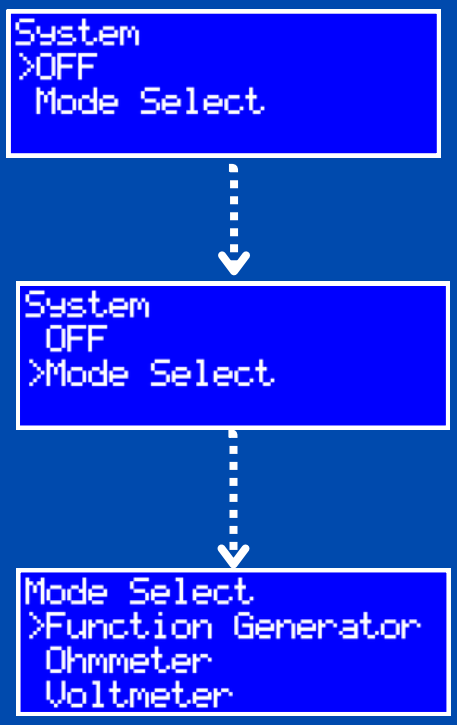
# POWER ON/OFF

WHEN YOU FIRST RUN THE ELECTRONICS TOOL BENCH SYSTEM, YOU ARE INTRODUCED TO A SYSTEM MENU PAGE. IT ALLOWS YOU TO TURN OFF THE SYSTEM, OR GO THROUGH MODE SELECT, WHICH ENABLES YOU TO SELECT AND INTERACT WITH THE DIFFERENT ELECTRONICS TOOLS. THE CONTROLS ARE EXPLAINED IN SECTION 1. ROTARY ENCODER.THE PARTICULARS OF EACH TOOL IS EXPLAINED IN THE SUBSEQUENT SECTIONS.

BOOTING THE SYSTEM BRINGS THIS SCREEN. THERE, YOU CAN TURN OFF THE SYSTEM BY PRESSING THE BUTTON.

SCROLL DOWN TO ‘MODE SELECT’ AND PRESS THE BUTTON TO START USING THE ELECTRONIC TOOL BENCH SYSTEM.

YOU ARE INTRODUCED TO A SCROLLABLE LIST OF TOOLS TO USE!




IN CASE YOU DO NOT KNOW THE CONTROLS, THE NEXT SECTION EXPLAINS ALL ABOUT THE ROTARY ENCODER!



# LIST OF COMPONENTS

# LIST OF COMPONENTS

## GENERAL PARTS

- PI HAT PROTECTOR (X1)
- PI HAT PROTECTOR FUSES (X2)
- 24V POWER SUPPLY (X1)
- LM339 QUAD COMPARATOR (X1)
- IRF4905 P CHANNEL MOSFET (X1)
- IRFZ34 N CHANNEL MOSFET (X1)

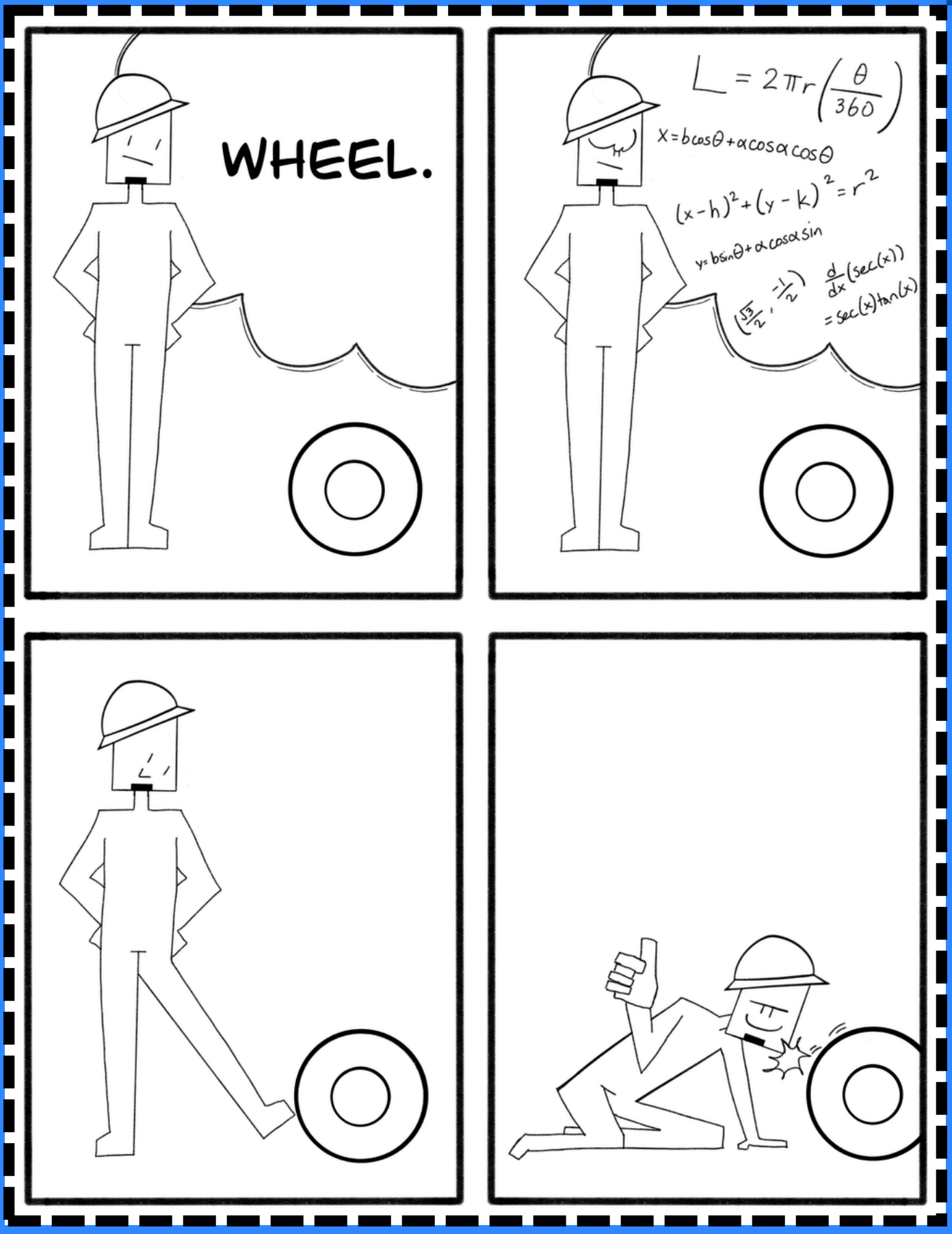
## TEST BENCH PARTS

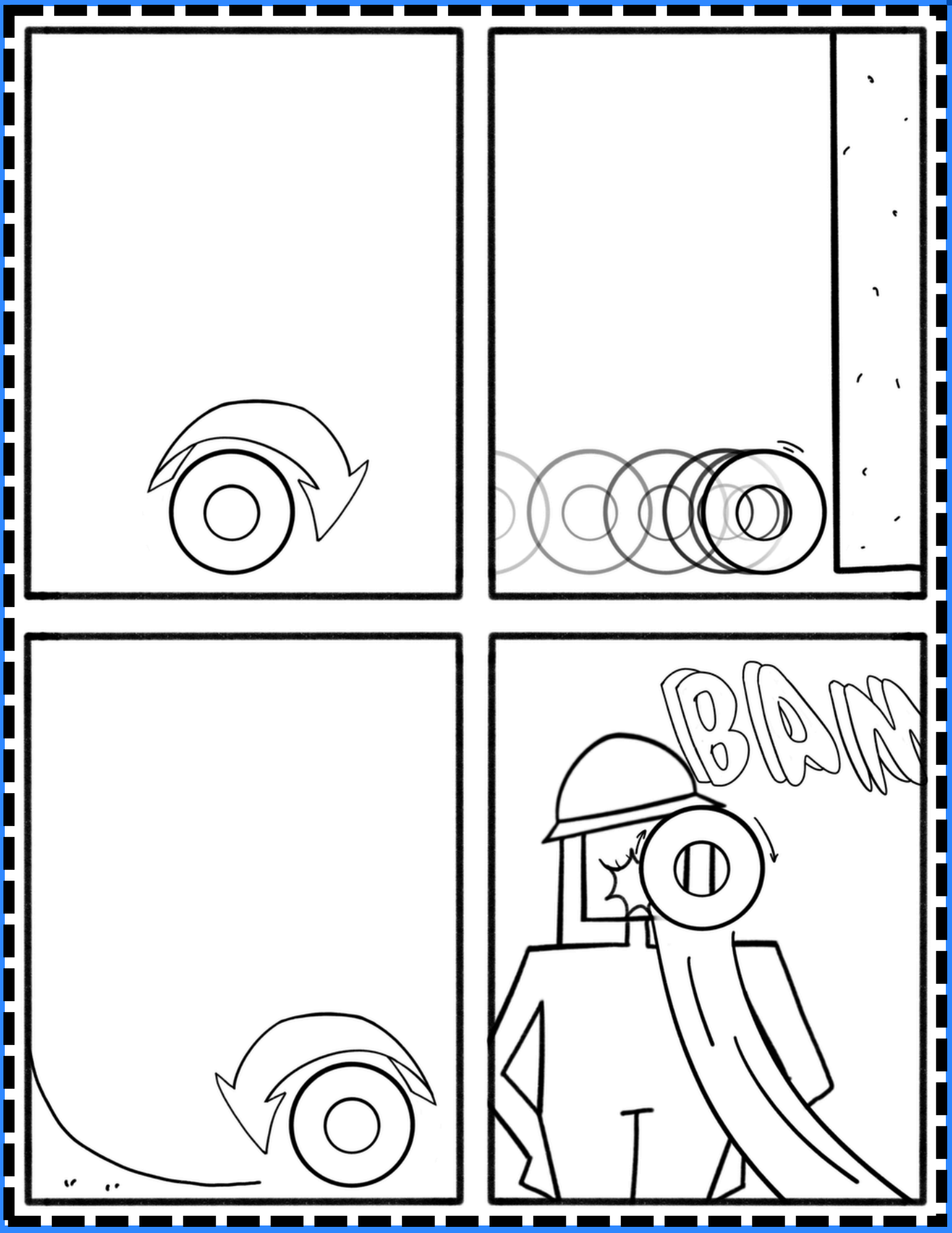
- MCP4131 SINGLE DIGIPOT (X2) OHMMETER, VOLTMETER
- MCP4231 DUAL DIGIPOT (X1) FUNCTION GENERATOR, DC REFERENCE
- 20X4 LCD (X1)
- KY-040 ROTARY ENCODER (X1)

## CANAKIT PARTS

- RASPBERRY PI 4 (X1)
- USB MICROSD CARD (X1)
- GPIO BREAKOUT WITH RIBBON CABLE (X1)

# 1. ROTARY ENCODER


WHEEL.
$L = 2\pi r\left(\frac{\theta}{360}\right)$
$x = b\cos\theta + a\cos\alpha\cos\theta$
$(x-h)^2 + (y-k)^2 = r^2$
$y = b\sin\theta + a\cos\alpha\sin$
$\left(\frac{\sqrt{3}}{2}, \frac{-1}{2}\right)$
$\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$


BAM

# ROTARY ENCODER

YOU CAN NAVIGATE AND INTERACT WITH THE USER INTERFACE ON THE LCD SCREEN BY TURNING OR PRESSING THE KNOB.

THERE ARE 3 CONTROLS THAT THE ELECTRONICS TOOL BENCH SYSTEM RESPONDS TO:

- KNOB TURN
- SHORT BUTTON PRESS
- LONG BUTTON HOLD

**⚠ CAUTION** DO NOT PULL ON THE ROTARY ENCODER KNOB.

**⚠ WARNING** DO NOT REMOVE THE ROTARY ENCODER WHILE THE ELECTRONIC TOOL BENCH SYSTEM IS ON.

## CONTROLS

YOU CAN TURN THE KNOB CLOCKWISE OR COUNTERCLOCKWISE FOR NAVIGATING A PAGE, OR ADJUSTING A VALUE (FOR DC REFERENCE VOLTAGE, WAVE FREQUENCY/AMPLITUDE).

ONLY SCROLLABLE PAGES, OR PAGES WHERE NOT ALL LINES ARE DISPLAYED ON THE LCD SCREEN AT ONCE, RESPOND FROM KNOB TURNS.

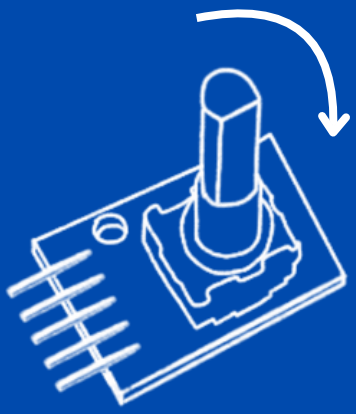
YOU CAN DO A SHORT BUTTON PRESS ON CONFIRMATION FOR A PAGE OPTION OR VALUE OF AN ADJUSTABLE MEASUREMENT.

NOTE: THE END OF A PAGE IS REPRESENTED BY THE LINES ‘BACK’ AND ‘MAIN’. PRESSING ‘BACK’ RETURNS TO THE PREVIOUS PAGE. PRESSING ‘MAIN’ RETURNS TO THE MAIN MENU.

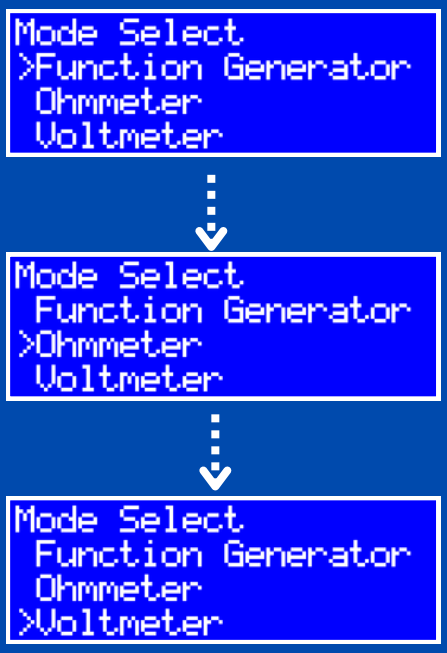
# PAGE NAVIGATION

TTHE FOLLOWING DEMONSTRATIONS SHOW THE BASIC CONTROLS FOR NAVIGATING THE ELECTRONIC TOOL BENCH.

**CLOCKWISE:**
**SCROLL DOWN**



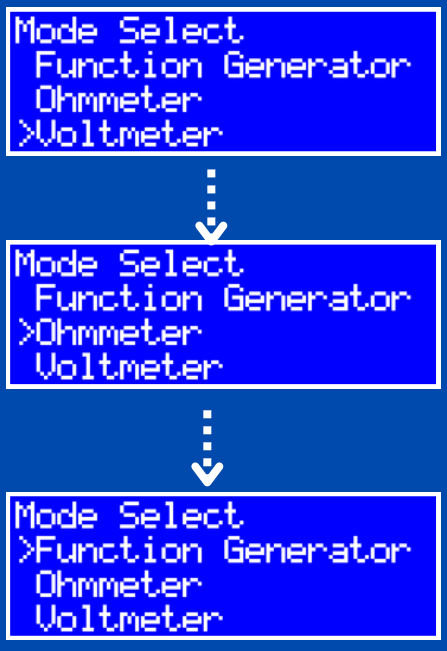
**DEMONSTRATION:**




**COUNTERCLOCKWISE:**
**SCROLL UP**

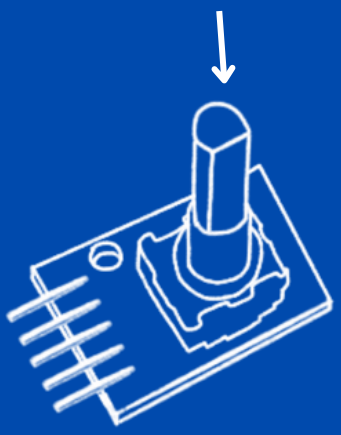


**DEMONSTRATION:**



# PAGE NAVIGATION (CONTINUED)

**BUTTON PRESS:**
**SELECT**



**DEMONSTRATION:**

```
Mode Select
>Function Generator
 Ohmmeter
 Voltmeter
```

(THE NEXT PAGE)

```
Function Generator
>Type
 Frequency
 Amplitude
```

IF YOU WANT TO RETURN TO THE PREVIOUS PAGE, PRESS ‘BACK’.
IF YOU WANT TO RETURN TO THE MAIN PAGE, PRESS ‘MAIN’.

## BACK OPTION DEMO

(START DEMO)

```
Function Generator
>Type
 Frequency
 Amplitude
```

SELECT

```
Type
>Sine
 Square
 Back
```

SCROLL DOWN

```
Type
 Sine
>Square
 Back
```

SCROLL DOWN

```
Type
 Sine
 Square
>Back
```

(THE PREVIOUS PAGE)

SELECT

```
Function Generator
>Type
 Frequency
 Amplitude
```

## MAIN OPTION DEMO

(START DEMO)

```
Function Generator
 Amplitude
>Output
 Back
```

SCROLL DOWN

```
Function Generator
 Output
>Back
 Main
```

SCROLL DOWN

```
Function Generator
 Output
 Back
>Main
```

(THE MAIN PAGE)

SELECT

```
Mode Select
>Function Generator
 Ohmmeter
 Voltmeter
```

# ADJUSTING MEASUREMENTS

YOU CAN ADJUST A MEASUREMENT TO THE DESIRED VALUE SHOWN ON THE LCD SCREEN. THIS IS APPLICABLE TO DC VOLTAGE REFERENCE AND FUNCTION GENERATOR.

YOU CAN INCREASE THE VALUE BY TURNING THE KNOB CLOCKWISE. YOU CAN DECREASE THE VALUE BY TURNING THE VALUE COUNTERCLOCKWISE.

A VALUE IS BOUNDED WITHIN A RANGE. YOU CANNOT INCREASE A VALUE ABOVE ITS MAXIMUM VALUE. YOU CANNOT DECREASE A VALUE BELOW ITS MINIMUM VALUE.

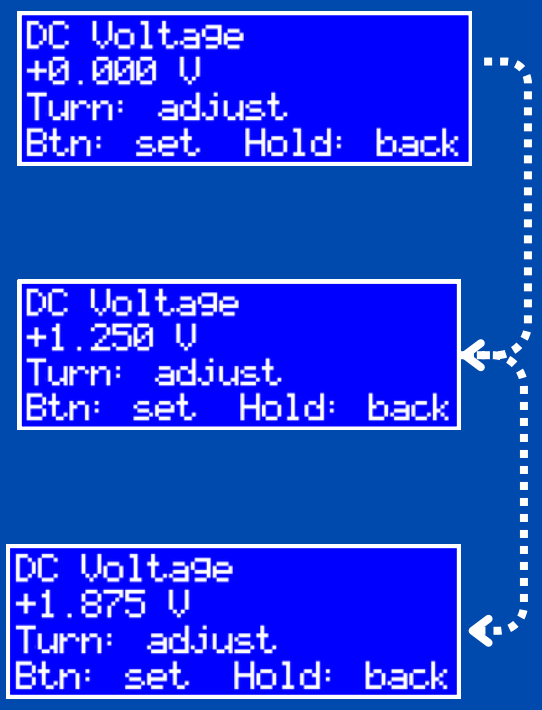
THESE ARE ALL ADJUSTABLE MEASUREMENTS THAT THE ELECTRONIC TOOL BENCH ALLOWS, WITH THE VALUE RANGE AND THE AMOUNT EACH TURN STEP CHANGES THE VALUE BY.

| Adjustable Measurement | Range with Units | Steps with Units |
|---|---|---|
| DC Voltage Reference | [-5, 5] V | 0.625 V |
| DC Voltage Reference | [100, 10,000]Hz | 500 Hz |
| Function Generator Amplitude | [-10, 10] V | 0.625 V |

ONCE YOU DECIDE THE VALUE TO SET AT, PRESS THE BUTTON.

**CLOCKWISE: INCREASE VALUE**

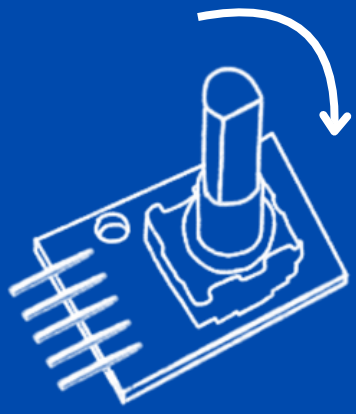


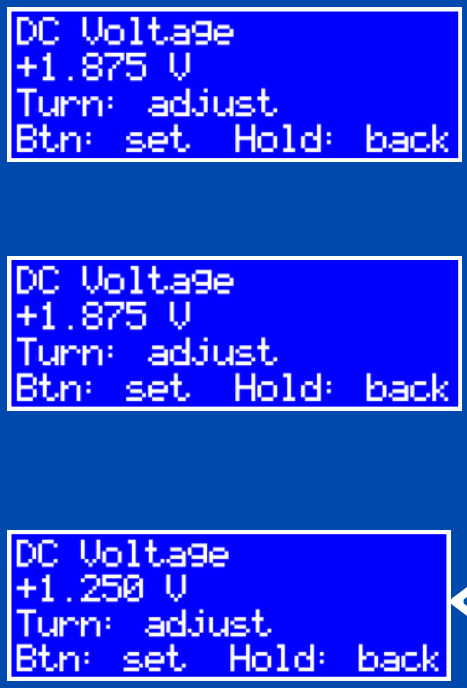


# ADJUSTING MEASUREMENTS (CONTINUED)

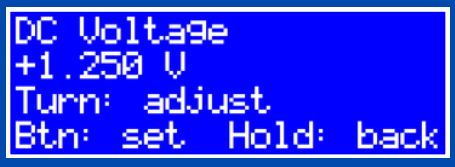
(PRESS BUTTON TO SELECT 1.875V)

**CLOCKWISE: DECREASE VALUE**



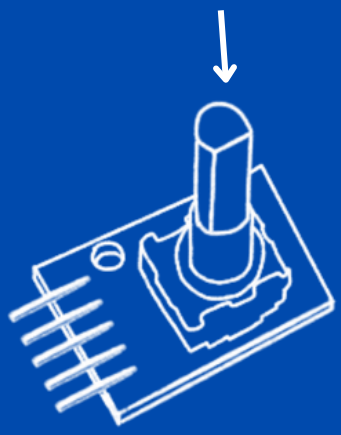


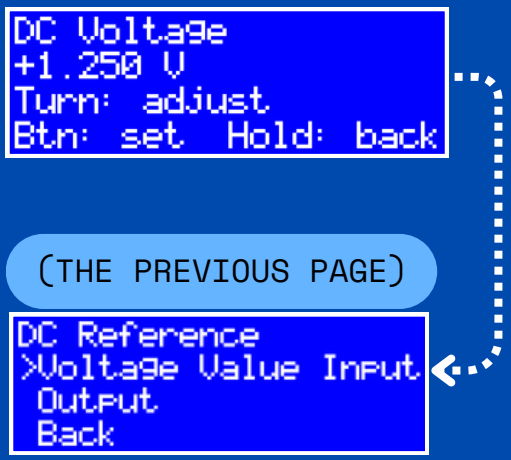

(PRESS BUTTON TO SELECT 1.250V)




YOU CAN DO A LONG BUTTON PRESS (A BUTTON HOLD THAT IS 3 SECONDS IN LENGTH) TO GO BACK TO THE PREVIOUS PAGE.

**BUTTON HOLD: 3 SECONDS**

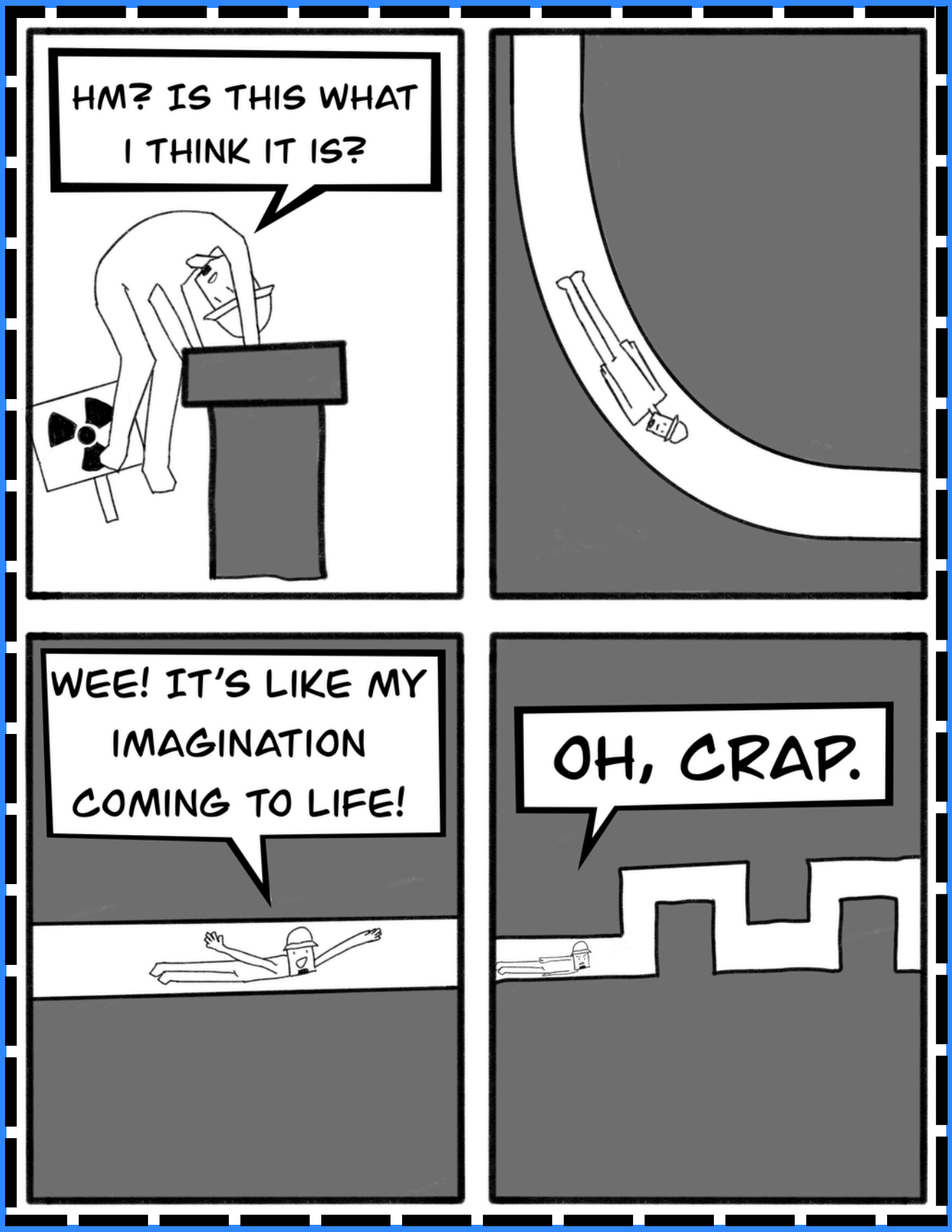




# 2. FUNCTION GENERATOR

2A. SQUARE WAVE GENERATOR
2B. SINE WAVE GENERATOR

# 2A.SQUARE WAVE GENERATOR


HM? IS THIS WHAT I THINK IT IS?
WEE! IT'S LIKE MY IMAGINATION COMING TO LIFE!
OH, CRAP.

# SQUARE WAVE GENERATOR

SUZIE, PLEASE HELP ME OUT!



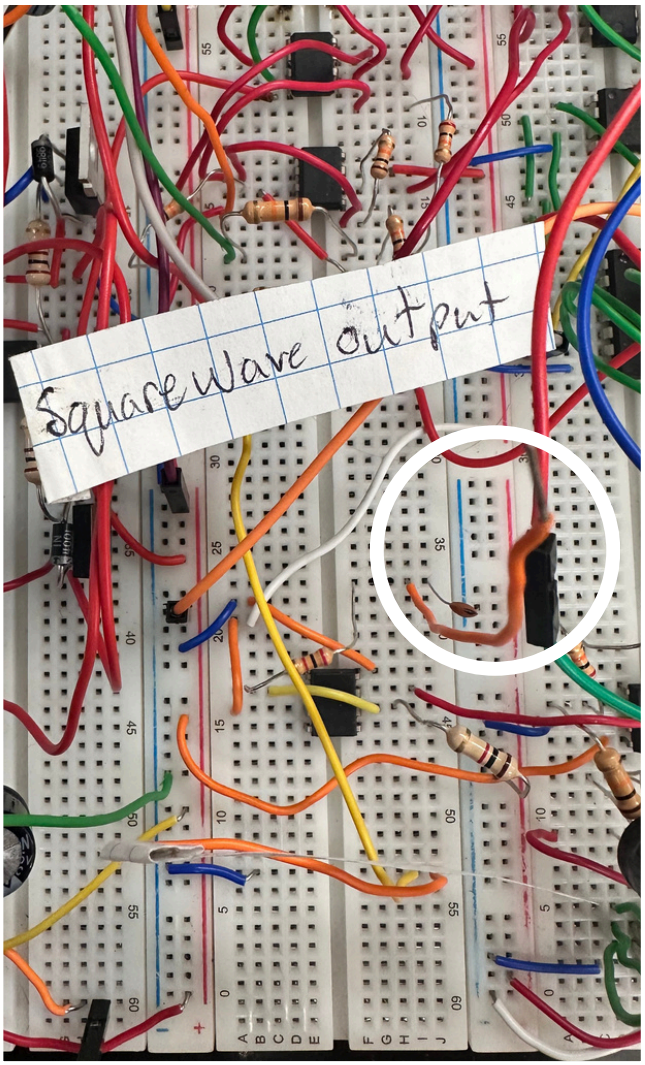
JAMAL WANTS A SQUARE WAVE WITH 1000HZ FREQUENCY AND 7.1875V AMPLITUDE. HOW DO WE HELP HIM?



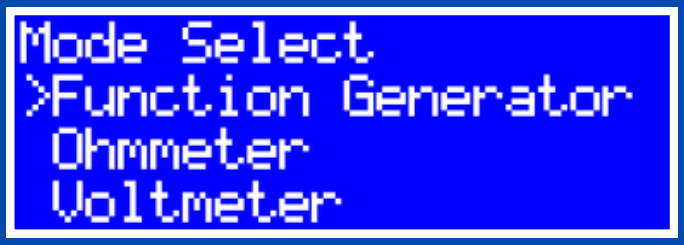
THIS SECTION AND THE FOLLOWING SECTIONS START WITH THE BREADBOARD CONNECTIONS YOU NEED TO MAKE. YOU SHOULD KNOW IF THE SYSTEM IS WORKING PROPERLY!



FIRST, ON THE BREADBOARD, IDENTIFY THE FLAG LABELED ‘SQUARE WAVE OUTPUT’. TO THE RIGHT OF THE FLAG, FIND AN UNCONNECTED ORANGE WIRE. THIS WIRE OUTPUTS THE SQUARE WAVE, AND IT IS CIRCLED FOR YOU.




SET THE WAVE TYPE TO SQUARE.






SCROLL TO ‘SQUARE’. PRESS THE BUTTON TO CONFIRM. TO GO BACK, PRESS THE BUTTON AGAIN.

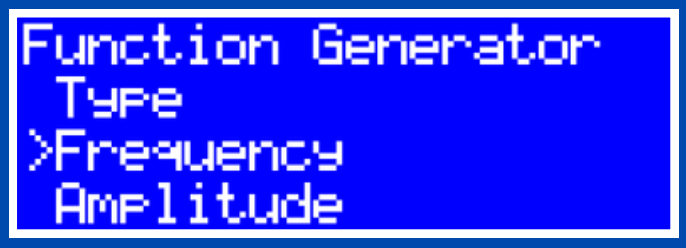






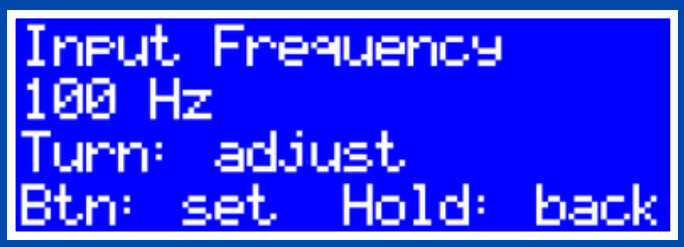



GO TO INPUT FREQUENCY.

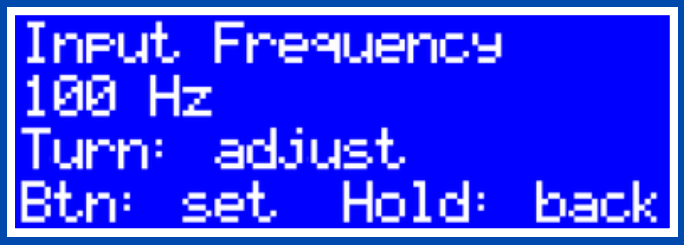


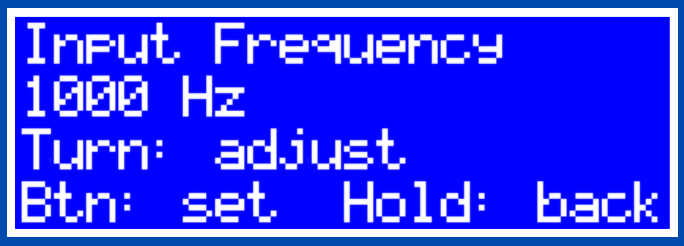






ADJUST TO 1000HZ. PRESS THE BUTTON TO SET THE VALUE. HOLD DOWN THE BUTTON TO GO BACK.



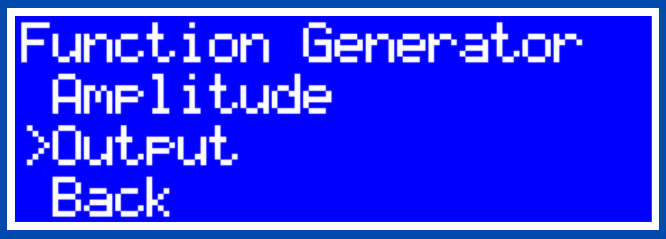

ADJUST TO 7.1875 V. PRESS THE BUTTON TO SET THE VALUE. HOLD DOWN THE BUTTON TO GO BACK.







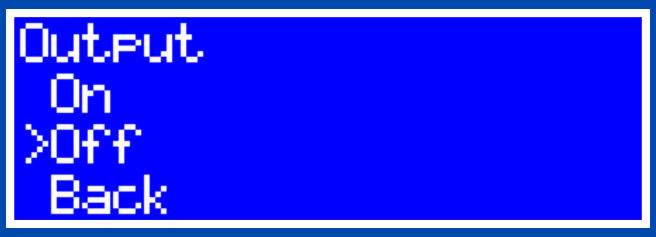
GO TO ‘OUTPUT’. TURN ON THE SQUARE WAVE.






CONGRATULATIONS! THE SQUARE WAVE OUTPUT IS ON!
OPTIONALLY, YOU CAN TURN IT OFF.



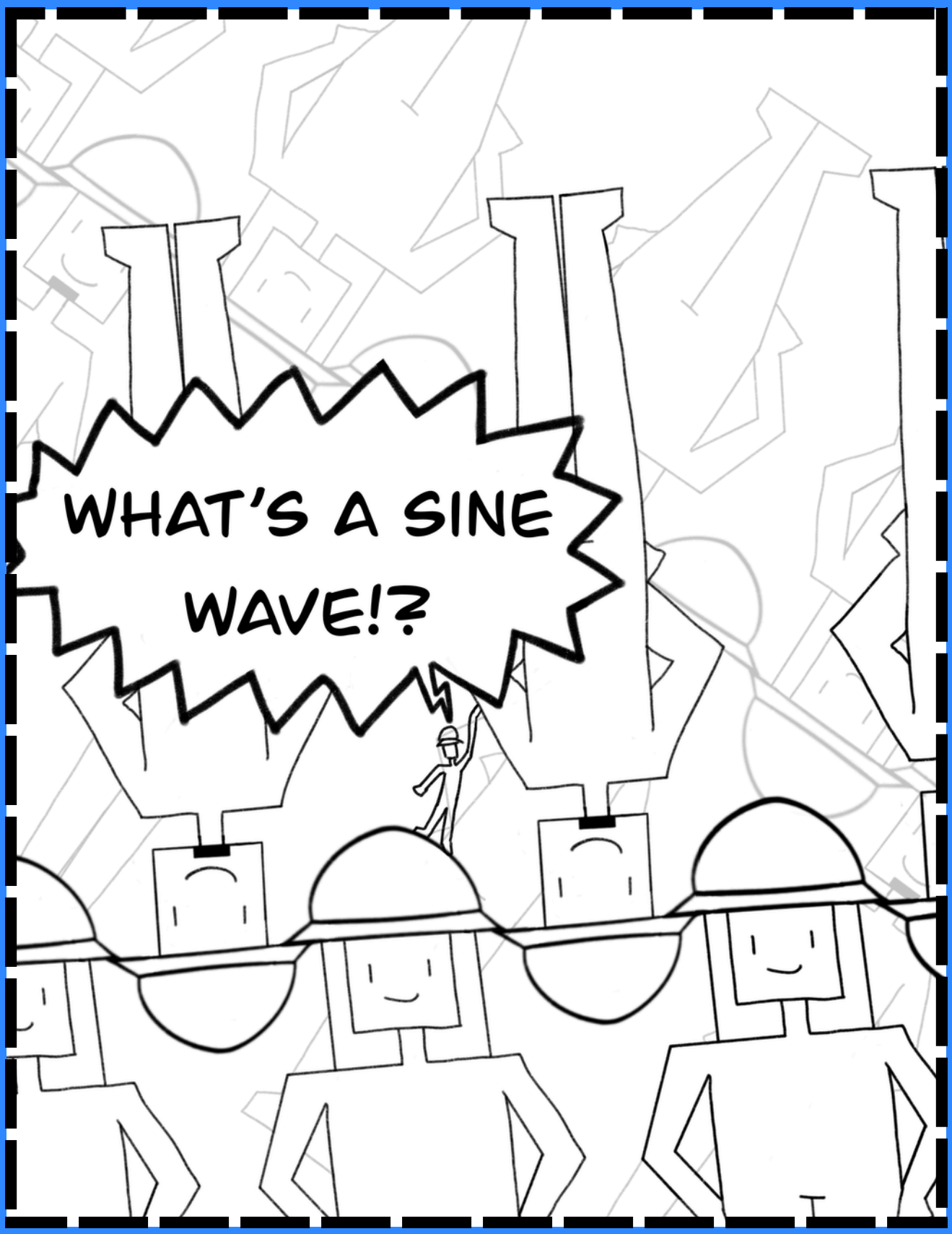




NOTE: IT IS OK TO IGNORE THE OFF OPTION. YOU CAN MAKE A NEW WAVE WITH A DIFFERENT FREQUENCY AND/OR AMPLITUDE AND THE NEW OUTPUT REPLACES THE OLD ONE.






# 2B.SINE WAVE GENERATOR


WHAT'S A SINE WAVE!?

# SINE WAVE GENERATOR

NOTE: LOOK OVER THE SQUARE WAVE GENERATOR SECTION FOR HELP SETTING FREQUENCY AND AMPLITUDE, AND TURNING ON/OFF THE OUTPUT WAVE.



MAKING A SINE WAVE IS AS EASY AS MAKING A SQUARE WAVE. IT IS THE SAME PROCESS!

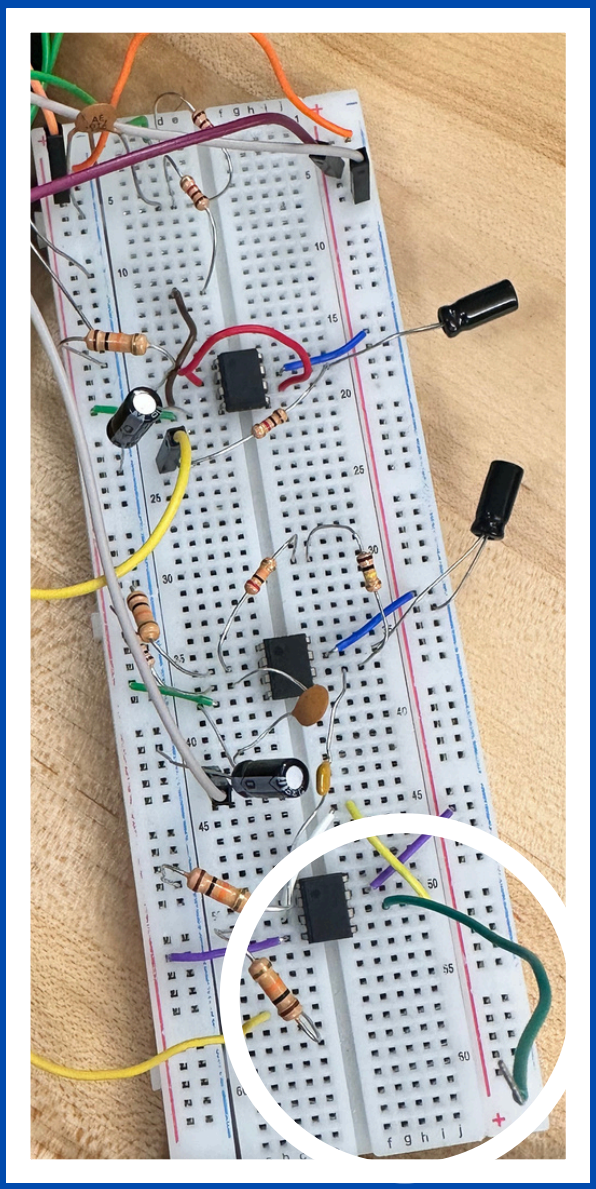


THE ONLY TWO DIFFERENCES ARE THAT YOU SELECT ‘SINE’ INSTEAD OF ‘SQUARE’...

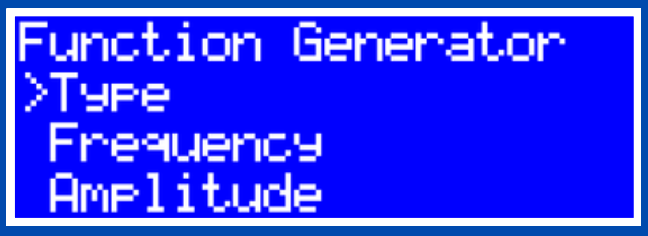


...AND THE WAVE OUTPUT IS IN A DIFFERENT PLACE.

LOOK TO THE RIGHT OF THE MAIN BREADBOARD. FIND AN UNCONNECTED GREEN WIRE. THIS WIRE OUTPUTS THE SINE WAVE, AND IT IS CIRCLED FOR YOU.



SELECT THE SINE OPTION UNDER TYPE.






AS BEFORE, PRESS THE BUTTON TO GO BACK.

```
Type:  Sine

Btn:  back
```

TURN ON THE OUTPUT. IF YOU SEE THIS SCREEN, THEN YOU DID IT. GREAT JOB!

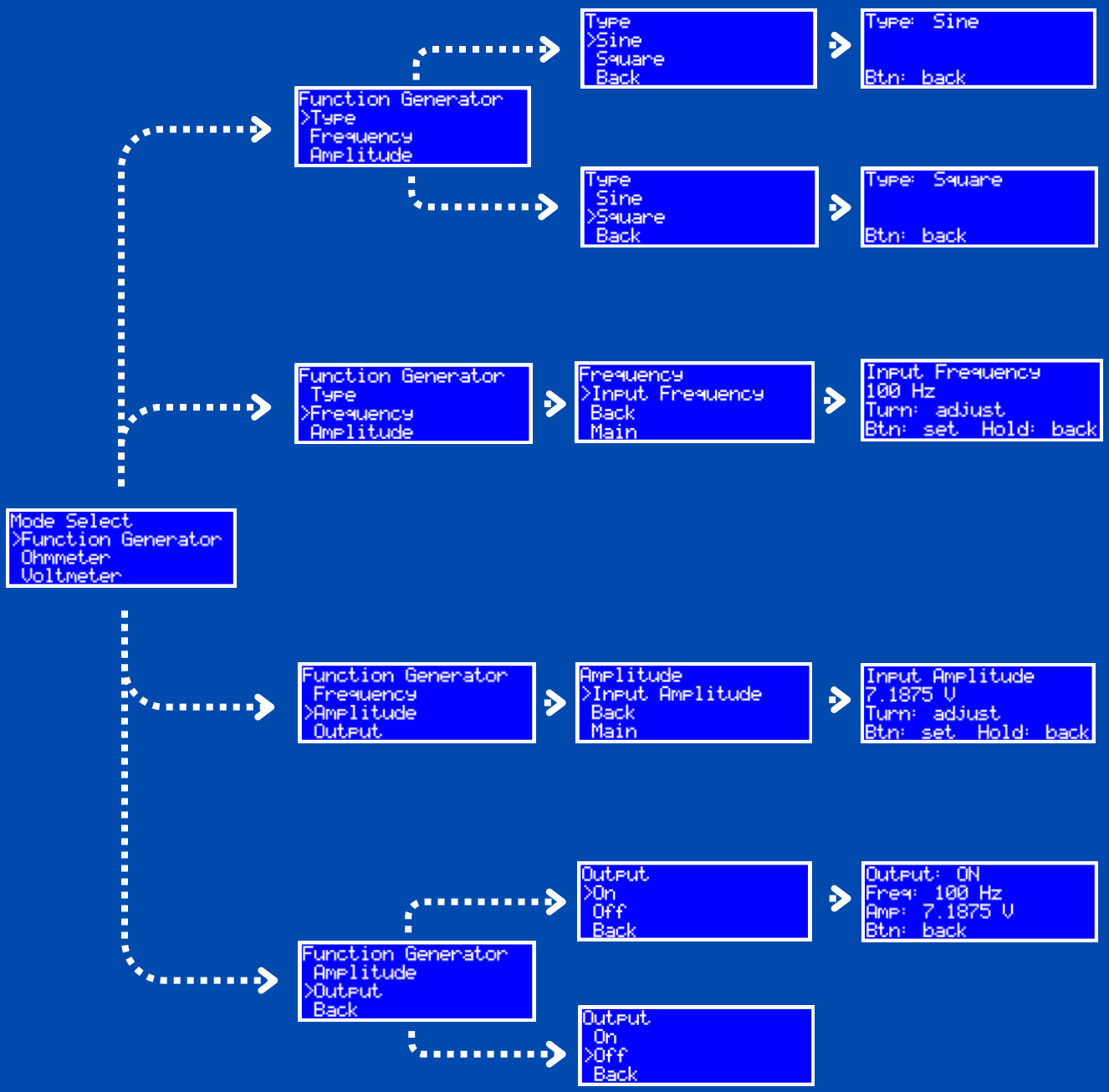


```
Output:  ON (Sine)
Freq:  10000 Hz
Amp: 5.0000 V
Btn:  back
```

AT THE END OF EACH ELECTRONIC TOOL SECTION, WE PROVIDE A ZOOM OUT OF THE UI: A UI DIAGRAM OVERVIEW. BIG PICTURE.



# 2. UI DIAGRAM OVERVIEW



# 3. OHMMETER


FREEZE!
PUT YOUR STATURE WHERE I CAN TAPE IT!
YOU'LL NEVER MEASURE ME ALIVE!
HE'S RESISTING!

# OHMMETER

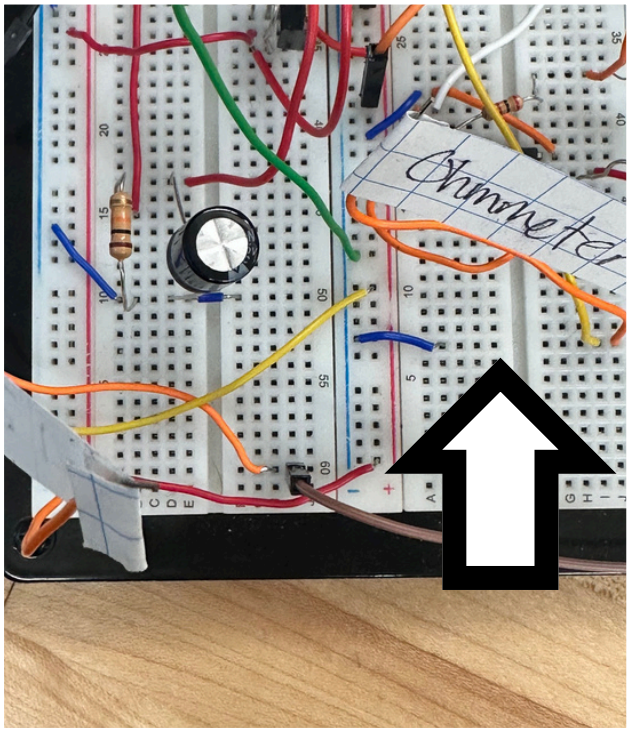
JAMAL IS BEHAVING LIKE A RESISTOR. LET'S MEASURE RESISTANCE!



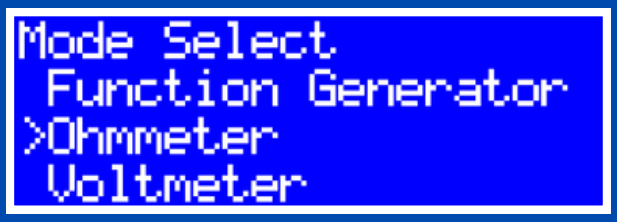
**⚠WARNING** DO NOT MEASURE YOURSELF WITH THE OHMMETER OR WITH ANY MEASURING TOOL.

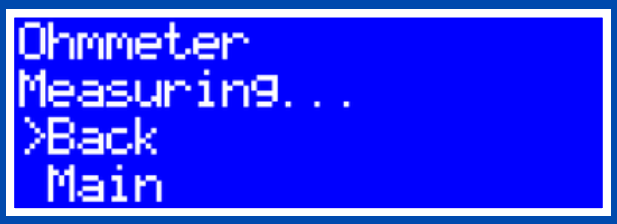
**⚠ CAUTION** BE CAREFUL TO CONNECT AN EXTERNAL RESISTOR TO THE CORRECT SPOT ON THE BREADBOARD.

AT THE BOTTOM LEFT CORNER OF THE BREADBOARD, IDENTIFY THE FLAG LABELED 'OHMMETER'. TO THE LEFT OF THE FLAG, FIND A GAP BETWEEN A BLUE WIRE AND A YELLOW WIRE. CONNECT THE RESISTOR TO MEASURE IN BETWEEN THE TWO WIRES.



SELECT THE OHMMETER. WAIT FOR IT TO FINISH MEASURING.

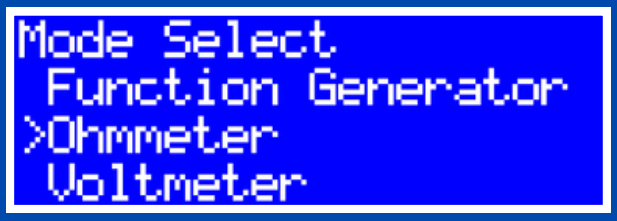


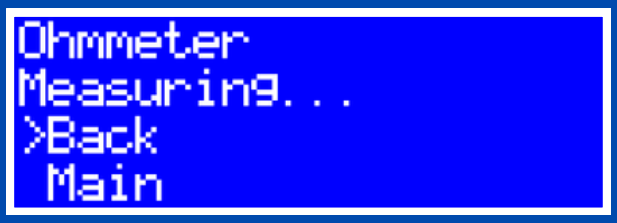




IF THE RANGE OF THE RESISTOR IS WITHIN [500, 10K]Ω, THE MEASURED AND TOLERANCE VALUES APPEAR. OTHERWISE, IT DISPLAYS 'NOT IN RANGE'.




GO BACK TO THE MAIN MENU. TAKE OUT THE RESISTOR AND MEASURE.



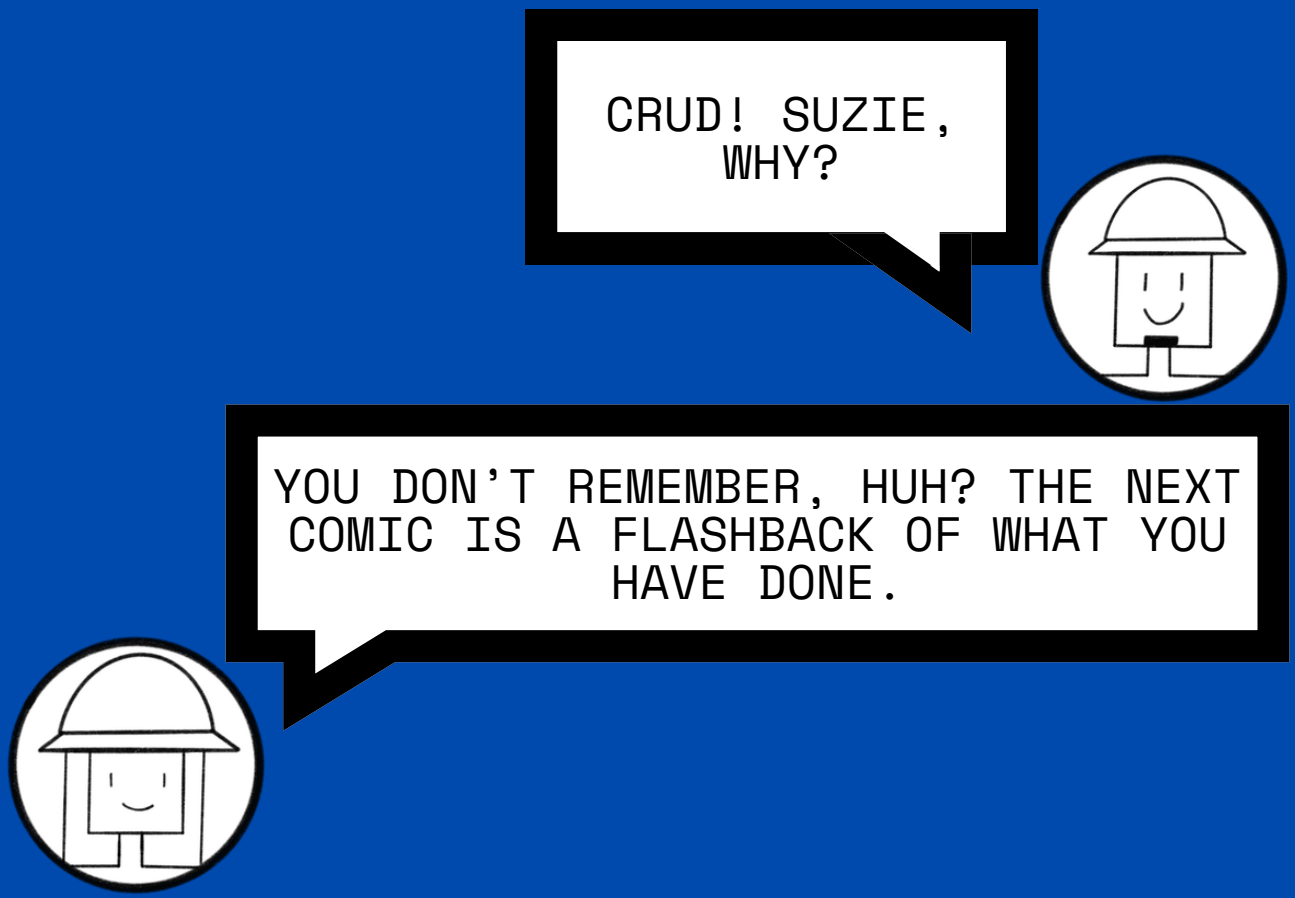

IT DISPLAYS 'NOT IN RANGE'. THE OHMMETER IS WORKING CORRECTLY.



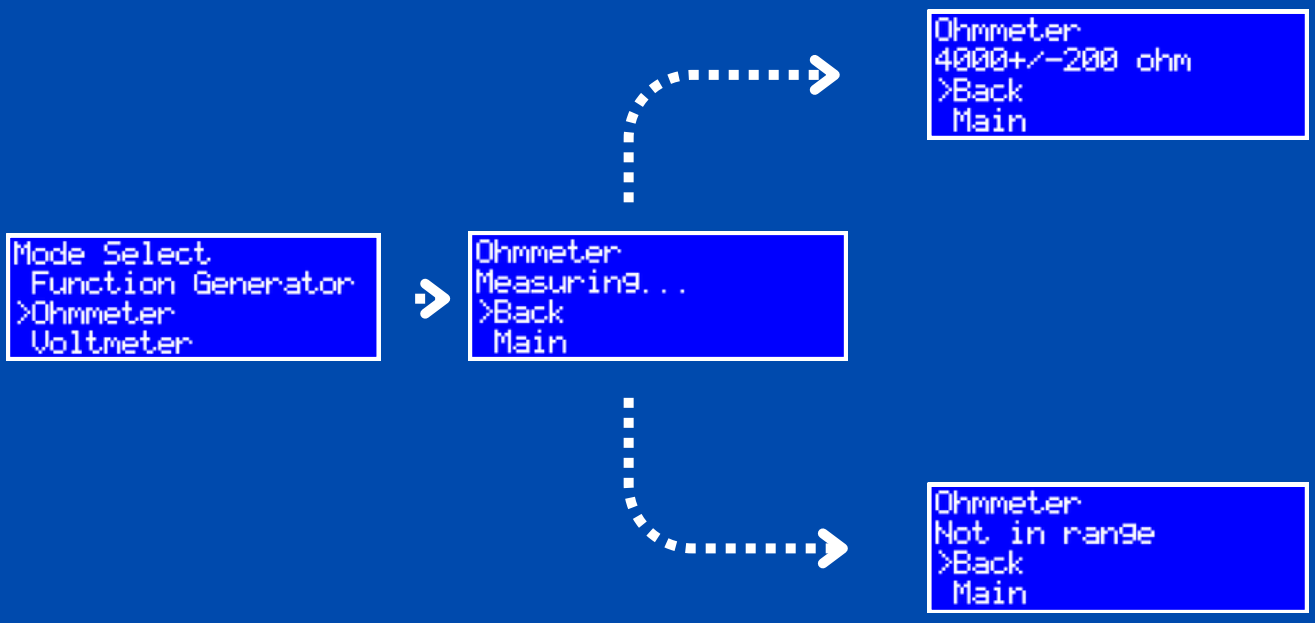
NOTE:

- AN OPEN CIRCUIT IS ONE WITH A DISCONNECTION,PREVENTING CURRENT FROM FLOWING. IDEALLY, THE RESISTANCE IS INFINITE.
- A SHORT CIRCUIT IS ONE WITH A WIRE ACTING AS A PATH OF LOW RESISTANCE. IDEALLY, THE RESISTANCE IS 0Ω.
- BOTH OPEN AND SHORT CIRCUITS ARE ‘OUT OF RANGE’.

**⚠WARNING** DO NOT MEASURE THE RESISTANCE OF ANY INTERNAL PART OF THE ELECTRIC TOOL BENCH. IT MAY DAMAGE IT OR SEND YOURSELF ELECTRIC SHOCK.



# 3. UI DIAGRAM OVERVIEW



# 4. VOLTMETER

4A. EXTERNAL
4B. INTERNAL REFERENCE


PANT
ANT
THIS CAMPING TRIP CAN'T GET ANY WORSE THAN THIS!
VOLKSWAGEN, MORE LIKE VOLTS-
THIS IS WHY I CAN'T PUT YOU IN CHARGE OF THINGS, JAMAL!

# 4A.EXTERNAL

# VOLTMETER - EXTERNAL

BUT THAT WAS AN ACCIDENT!



AN EXPENSIVE ONE AT THAT!



WE HAVE TO MOVE ON, LET'S MEASURE VOLTAGE.

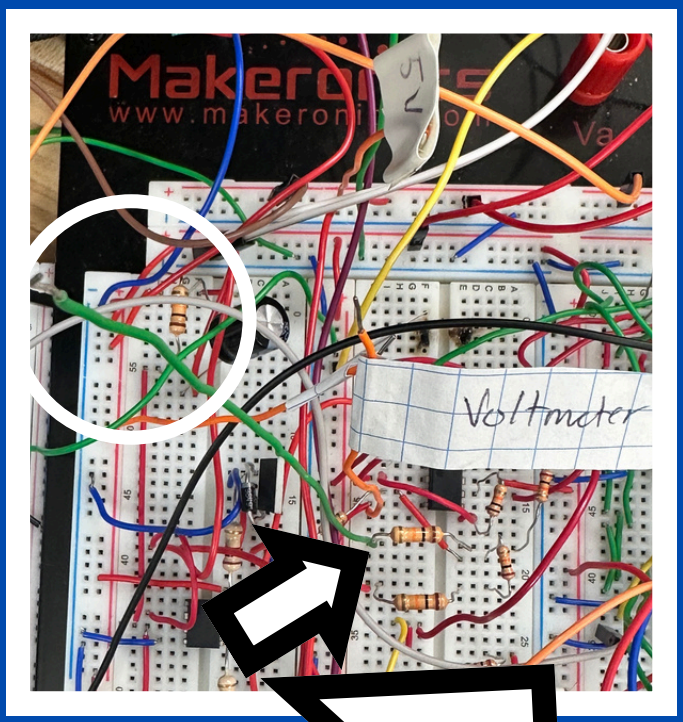


**⚠WARNING** BEFORE USE, CONFIRM THAT THE VOLTMETER TOOL IS CONNECTED TO GROUND.

**⚠WARNING** BE CAREFUL TO CONNECT AN EXTERNAL VOLTAGE SOURCE TO THE CORRECT SPOT ON THE BREADBOARD.

**⚠WARNING** IF YOU ARE UNSURE ABOUT THE DESIGN OF THE ELECTRONICS TOOL BENCH SYSTEM, DO NOT CONNECT GROUND TO ANY OTHER RAIL.

AT THE TOP LEFT CORNER OF THE BREADBOARD, IDENTIFY THE FLAG LABELED ‘VOLTMETER’. BELOW THE FLAG, FIND AN UNCONNECTED GREEN WIRE. IT IS CIRCLED FOR YOU.




USE TWO PROBES TO MAKE THE CONNECTION: ONE FOR INPUT VOLTAGE, AND ANOTHER FOR GROUND. CONNECT THE INPUT VOLTAGE TO THE GREEN WIRE. CONNECT THE GROUND TO THE BREADBOARD AS INDICATED BY THE BIG ARROW.

GROUND RAIL IS THIS BLUE LINE

TO MEASURE AN EXTERNAL VOLTAGE SOURCE, SELECT ‘EXTERNAL’.

```
Mode Select
 Function Generator
 Ohmmeter
>Voltmeter
```

```
 Source
>External
 Internal Reference
 Back
```

IF THE RANGE OF THE VOLTAGE IS WITHIN [-5, 5]V, THE MEASURED AND TOLERANCE VALUES APPEAR. OTHERWISE, IT DISPLAYS ‘NOT IN RANGE’.

```
Voltmeter
Measuring...
>Back
 Main
```

```
Voltmeter
+0.00V +/-0.15V
>Back
 Main
```

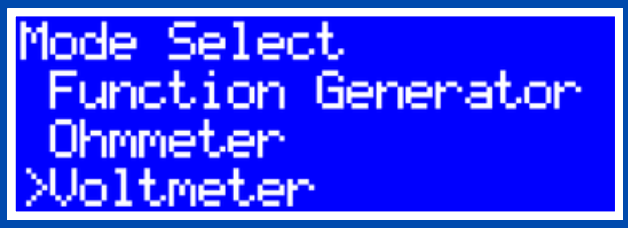


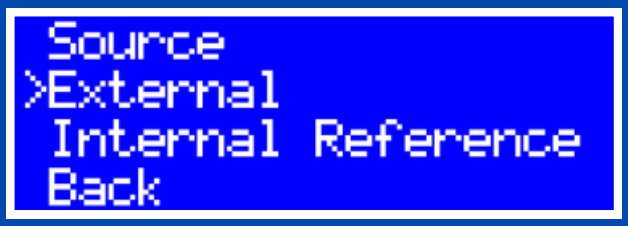
AS YOU CAN SEE, THE USER INTERFACE IS SIMILAR TO THAT OF THE OHMMETER.

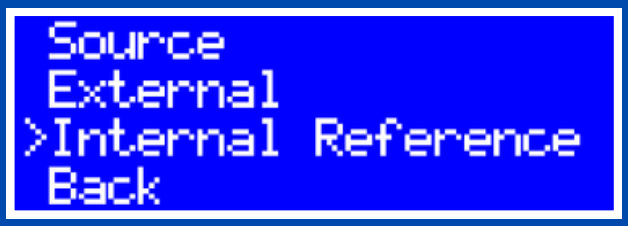
# 4B. INTERNAL REFERENCE

# VOLTMETER - INTERNAL REFERENCE

TO MEASURE AN INTERNAL VOLTAGE SOURCE, SELECT ‘INTERNAL REFERENCE’.

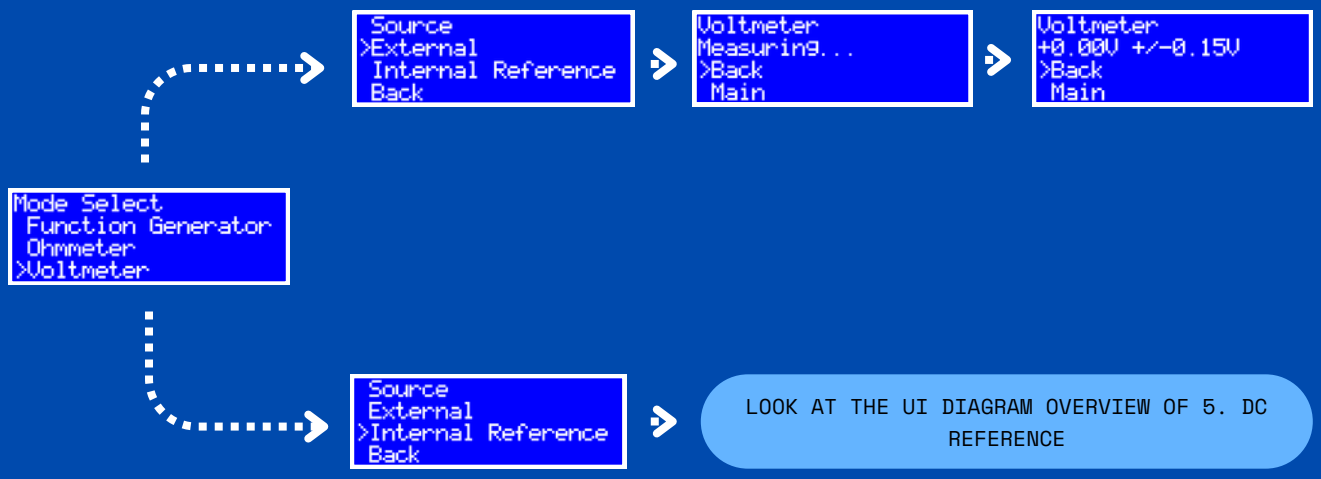


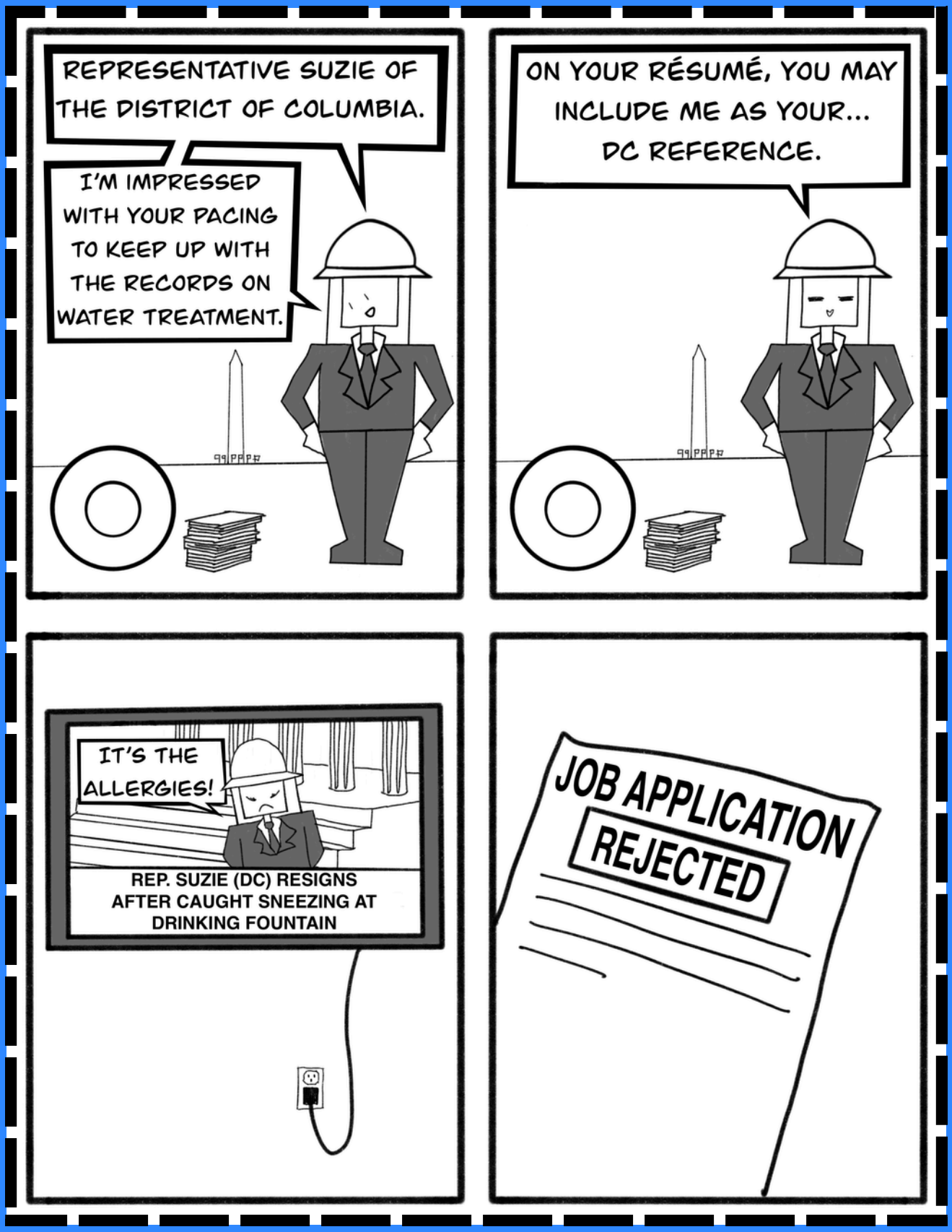




THE INTERNAL REFERENCE IS EQUIVALENT TO THE USER INTERFACE FOR IN 5. DC REFERENCE.

# 4. UI DIAGRAM OVERVIEW



# 5. DC REFERENCE


REPRESENTATIVE SUZIE OF THE DISTRICT OF COLUMBIA.
I'M IMPRESSED WITH YOUR PACING TO KEEP UP WITH THE RECORDS ON WATER TREATMENT.
ON YOUR RÉSUMÉ, YOU MAY INCLUDE ME AS YOUR... DC REFERENCE.
IT'S THE ALLERGIES!
REP. SUZIE (DC) RESIGNS AFTER CAUGHT SNEEZING AT DRINKING FOUNTAIN
JOB APPLICATION
REJECTED

# DC REFERENCE



WE CAN MEASURE VOLTAGE. BUT WHAT IF WE CAN ALSO SUPPLY VOLTAGE?

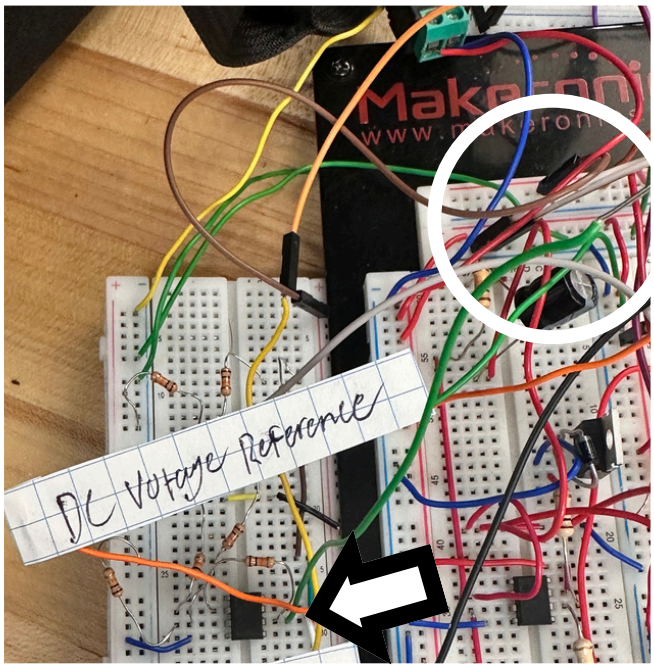
GIVE ME 5 VOLTS, SUZIE!



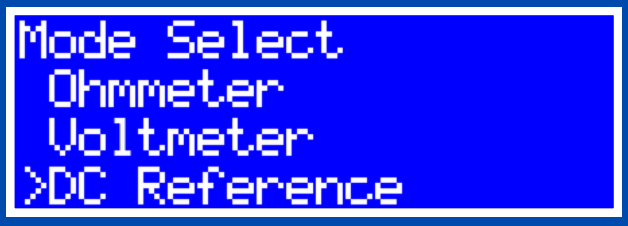
BE PATIENT, JAMAL. WHY DON'T WE LEARN THIS TOGETHER?



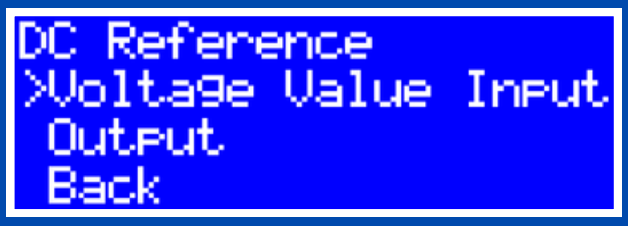
AT THE UPPER HALF OF THE LEFT BREADBOARD, IDENTIFY THE FLAG LABELED 'DC VOLTAGE REFERENCE'. TO THE LEFT OF THE FLAG, FIND AN UNCONNECTED GREEN WIRE. THIS WIRE OUTPUTS A VOLTAGE REFERENCE, AND IT IS CIRCLED FOR YOU.



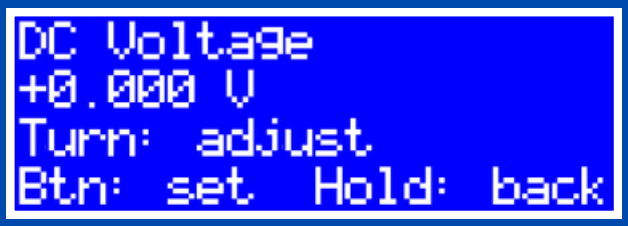
SELECT THE DC REFERENCE.

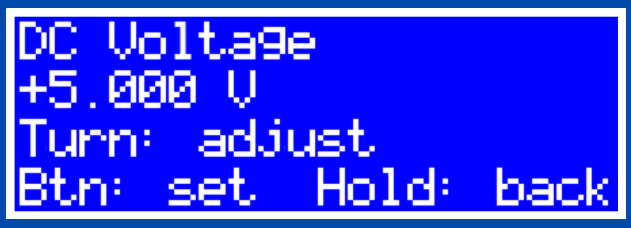



START BY ADJUSTING DC INPUT.




ADJUST TO 5V. PRESS THE BUTTON TO SET THE VALUE. HOLD DOWN THE BUTTON TO GO BACK.




SELECT ‘OUTPUT’ TO SWITCH THE OUTPUT ON/OFF.




NOTE: IF YOU SELECT ‘BACK’ OR ‘MAIN’ FROM THE DC REFERENCE PAGE, THEN THE DC OUTPUT IS TURNED OFF.

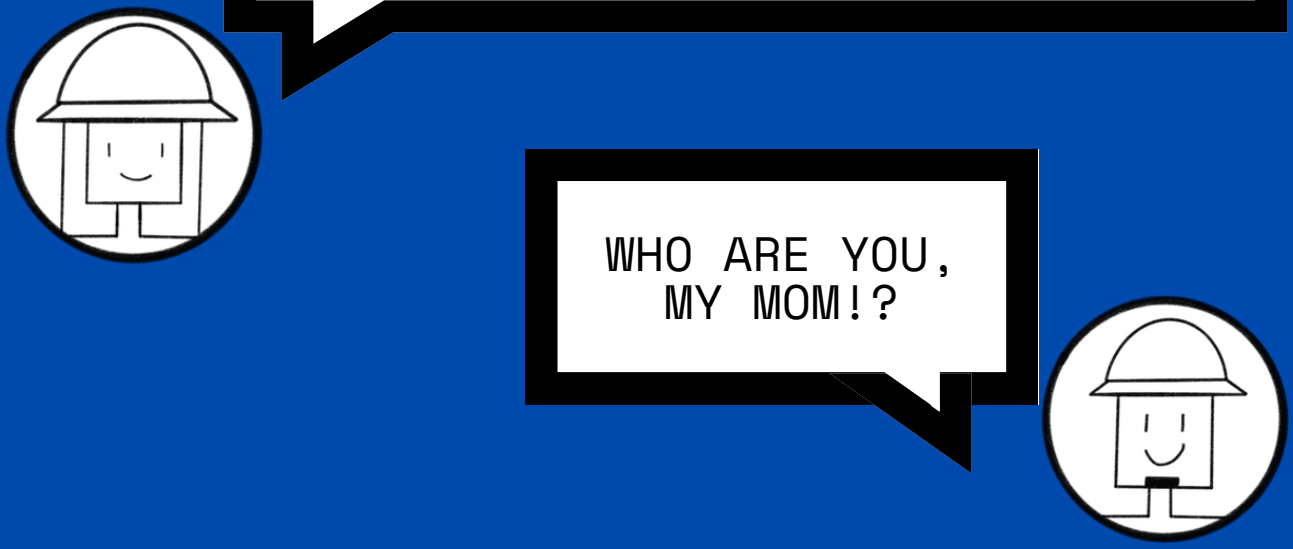
TURN ON THE DC OUTPUT. UPON TURNING IT ON, YOU SEE THE SET AND MEASURED OUTPUT VOLTAGES. PRESS THE BUTTON AGAIN.

```
DC Output
>On
 Off
 Back
```

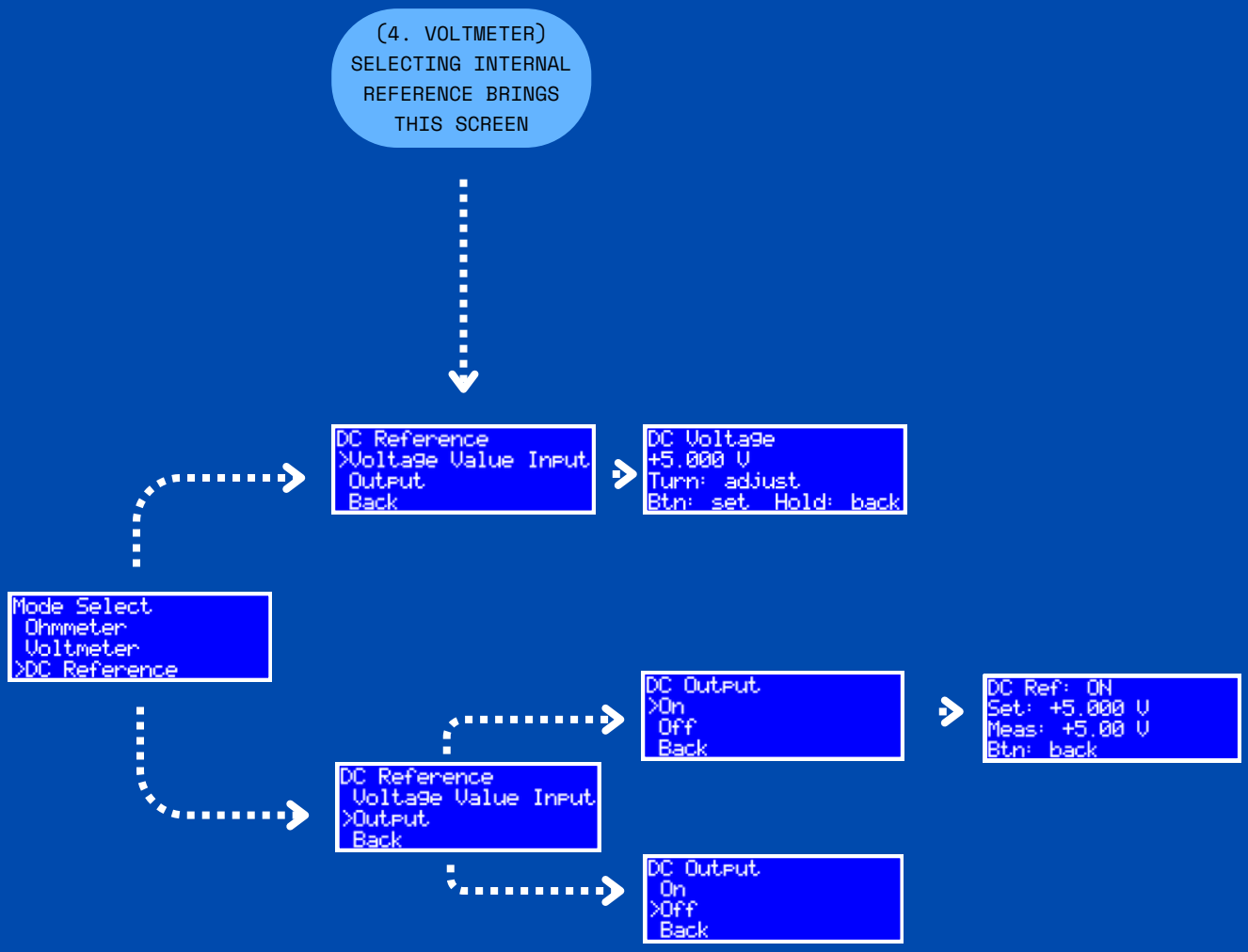
```
DC Ref: ON
Set: +5.000 V
Meas: +5.00 V
Btn: back
```

OPTIONALLY, YOU CAN TURN IT OFF.

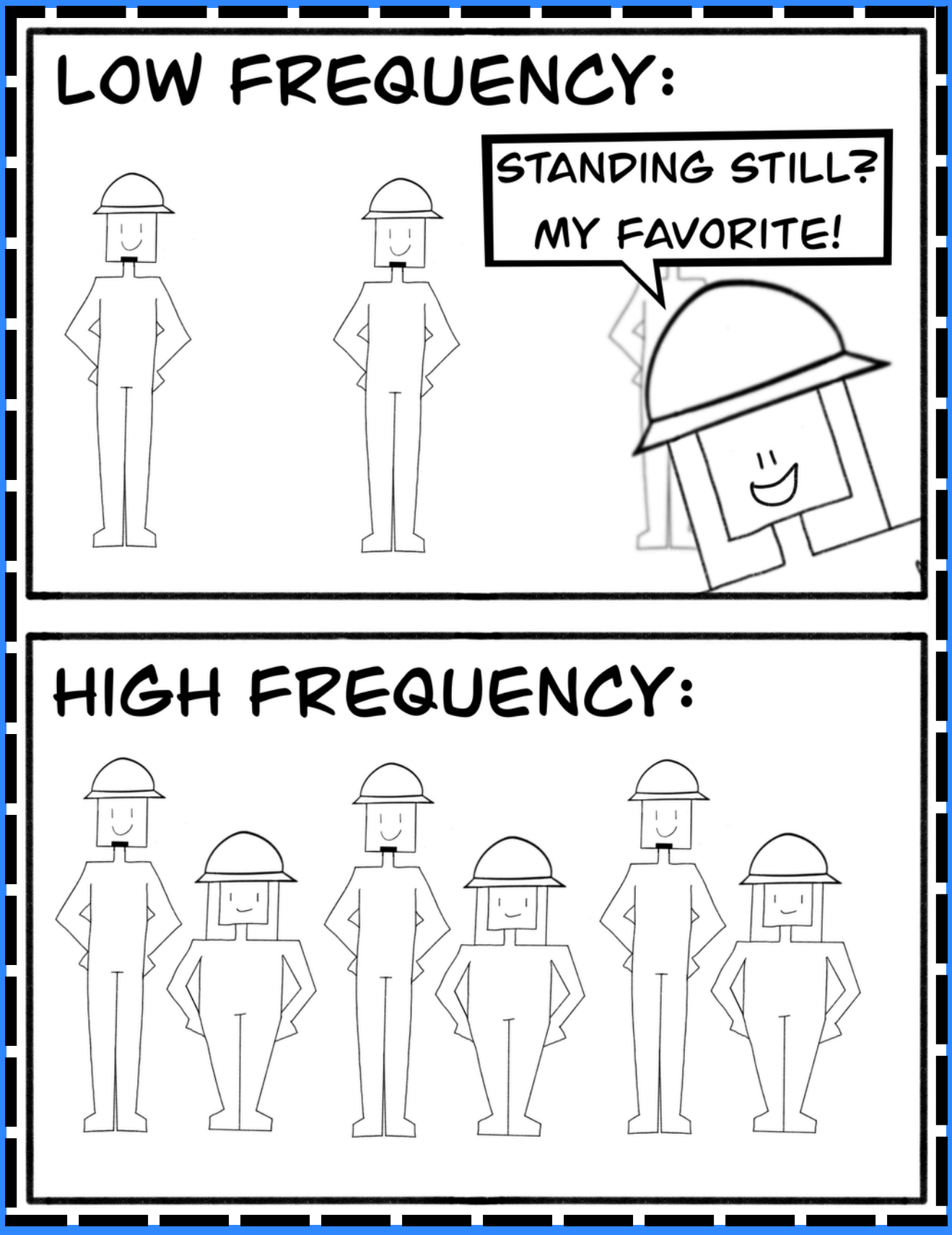
```
DC Output
 On
>Off
 Back
```



# 5. UI DIAGRAM OVERVIEW

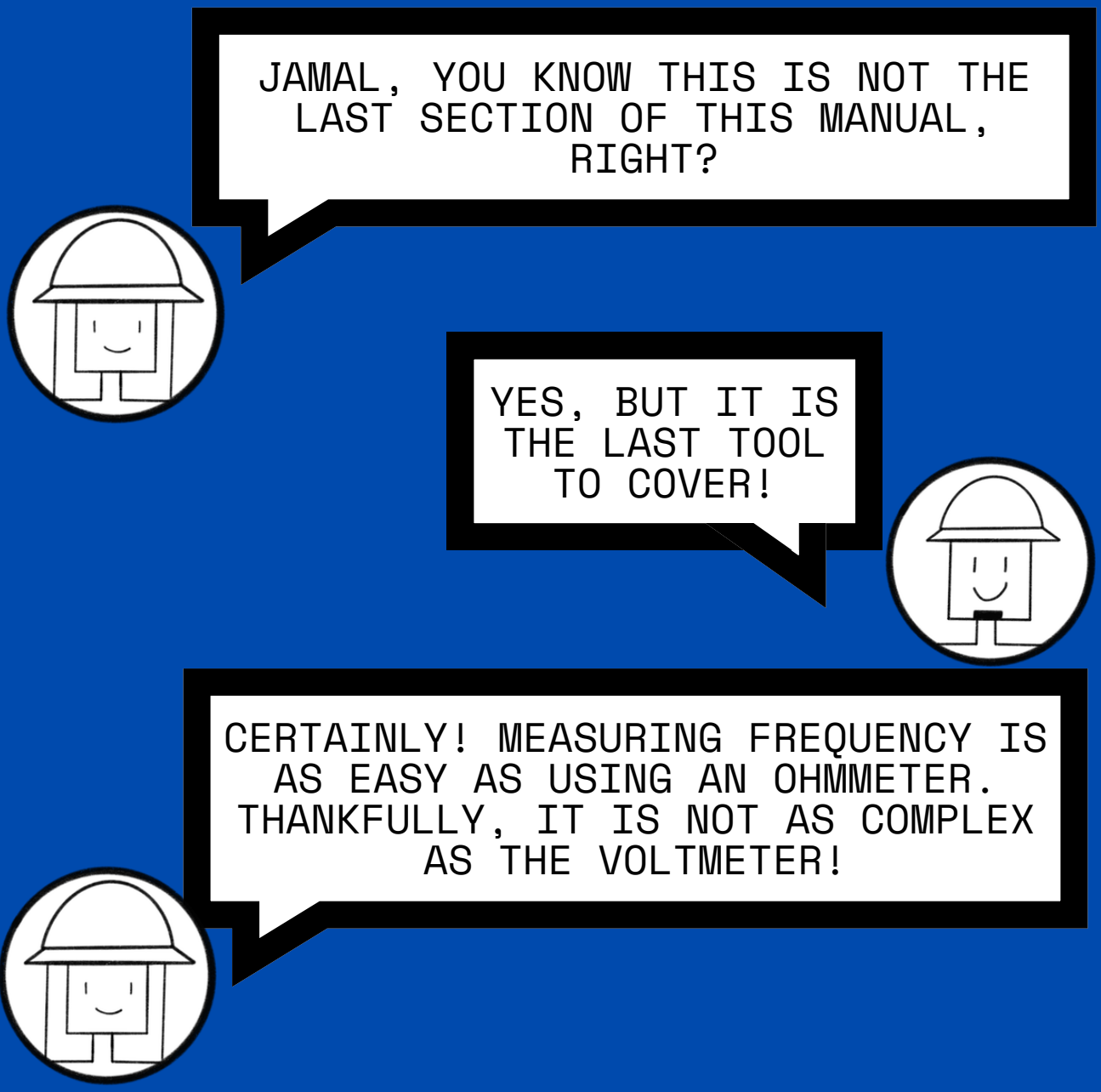


# 6. FREQUENCY MEASUREMENT


LOW FREQUENCY:
STANDING STILL? MY FAVORITE!
HIGH FREQUENCY:

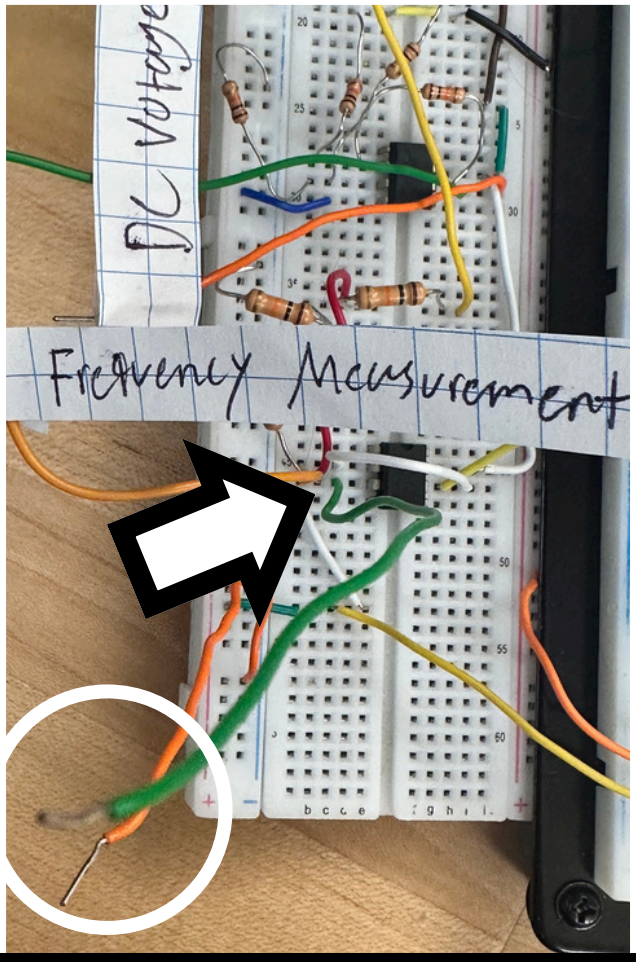
# FREQUENCY MEASUREMENT

NOTE: THE INPUT MUST BE A SINE WAVE WITH A FREQUENCY VALUE BETWEEN 1KHZ AND 10KHZ.



**⚠ CAUTION** THE FREQUENCY MEASUREMENT IS DESIGNED FOR SINE WAVES ONLY. USING ANY OTHER KIND OF WAVE MAY GIVE INCORRECT MEASUREMENTS.

AT THE LOWER HALF OF THE LEFT BREADBOARD, IDENTIFY THE FLAG LABELED ‘FREQUENCY MEASUREMENT’. TO THE RIGHT OF THE FLAG, FIND AN UNCONNECTED GREEN WIRE. THIS WIRE CONNECTS THE INPUT SINE WAVE, AND IT IS CIRCLED FOR YOU.




THE GREEN WIRE, NOT THE ORANGE WIRE! JAMAL WOULD MAKE THIS KIND OF MISTAKE...



TO MEASURE FREQUENCY, SELECT ‘FREQUENCY MEAS.’

```
Mode Select
 Voltmeter
 DC Reference
>Frequency Meas.
```

WAIT FOR THE SYSTEM TO COMPLETE MEASURING.

```
Freq Measurement
Waiting...
>Back
 Main
```

THE MEASUREMENT IS COMPLETE AND DISPLAYS THE FREQUENCY VALUE.

```
Freq Measurement
8450.1 Hz
>Back
 Main
```

IF THE FREQUENCY IS OUT OF RANGE, THEN THE SCREEN SHOWS ‘NO SIGNAL’.

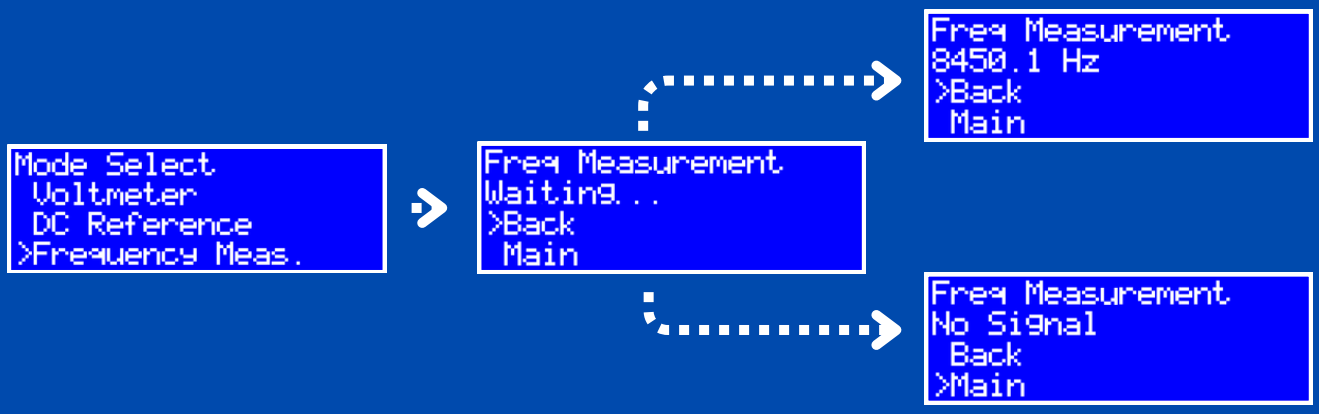
```
Freq Measurement
No Signal
 Back
>Main
```



WHY ARE THERE SO MANY NOTES? I’M TINY!

NOTE: THE WAITING TIME MAY DEPEND ON THE FREQUENCY VALUE OF THE SINE WAVE YOU ARE MEASURING. TYPICALLY, INPUTS WITH LOWER FREQUENCIES REQUIRE LONGER WAIT TIMES BECAUSE THEY NEED A HIGHER RESOLUTION FOR HIGHER ACCURACY.

# 6. UI DIAGRAM OVERVIEW



# FAQ

# FAQ

I TURNED ON THE LCD, BUT WHY IS THE ROTARY ENCODER NOT WORKING?



MAKE SURE THE 24V DC POWER SUPPLY IS CONNECTED TO THE SYSTEM. IF YOU ARE UNSURE, GO TO PG. 8.



HOW DO I TURN OFF THE ELECTRONICS TOOL BENCH SYSTEM?



ON THE MAIN MENU, SELECT ‘BACK’. YOU CAN TURN OFF THE SYSTEM BY SELECTING ‘OFF’.



I NEED TO REPLACE A SPECIFIC COMPONENT ON THE SYSTEM. HOW DO I KNOW WHAT SPECIFIC PART TO REPLACE?



RESORT TO THE LIST OF COMPONENTS ON PG. 10. TO IDENTIFY IF THE DAMAGED PART IS LISTED.



THE SYSTEM FROZE SUDDENLY! WHAT DO I DO?



UNPLUG THE POWER SUPPLY AND PLUG IT BACK IN TO RESET THE PROGRAM.



WHILE I AM MEASURING, KNOWING I MADE THE RIGHT CONNECTIONS, WHY AM I NOT SEEING A VALUE?



THE INPUT COULD BE OUT OF RANGE. GO TO THE SECTION ‘ADJUSTING MEASUREMENTS’ ON PG. 18 FOR THE TABLE CONTAINING THE ACCEPTABLE RANGES OF INPUT.



DOES THE FREQUENCY MEASUREMENT WORK FOR SQUARE WAVES, TOO?



THE TOOL WILL NOT BE AS ACCURATE BECAUSE IT IS DESIGNED SPECIFICALLY FOR SINE WAVES.

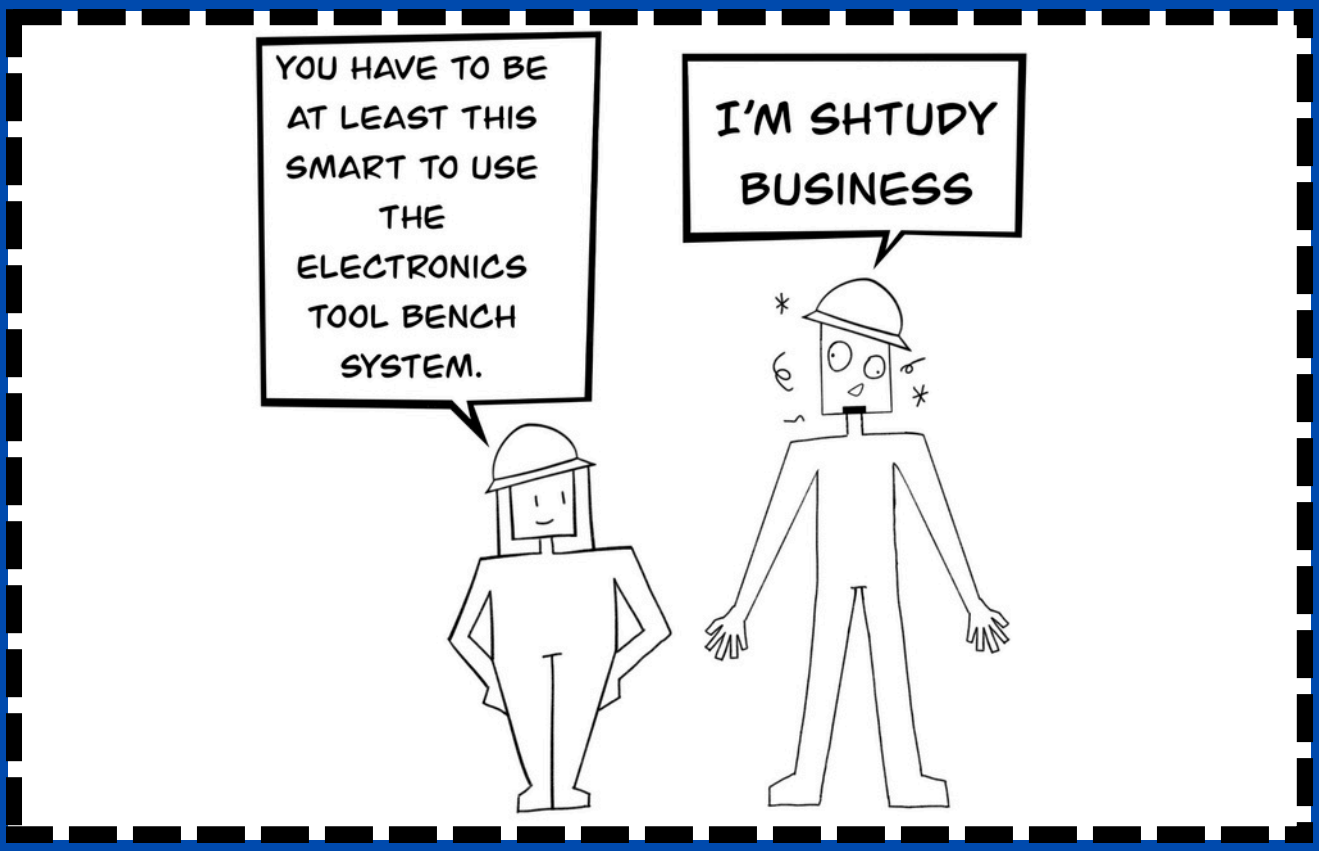


IT IS WEIRD THAT I GOT HIT BY THAT WRENCH EARLIER. HOW AM I SPEAKING SO COHERENTLY?



THIS WAS PAGES AGO... THE BACK COVER SHOWS WHAT HAPPENED RIGHT AFTER THAT INCIDENT...




YOU HAVE TO BE AT LEAST THIS SMART TO USE THE ELECTRONICS TOOL BENCH SYSTEM.
I'M SHTUDY BUSINESS

Group 5